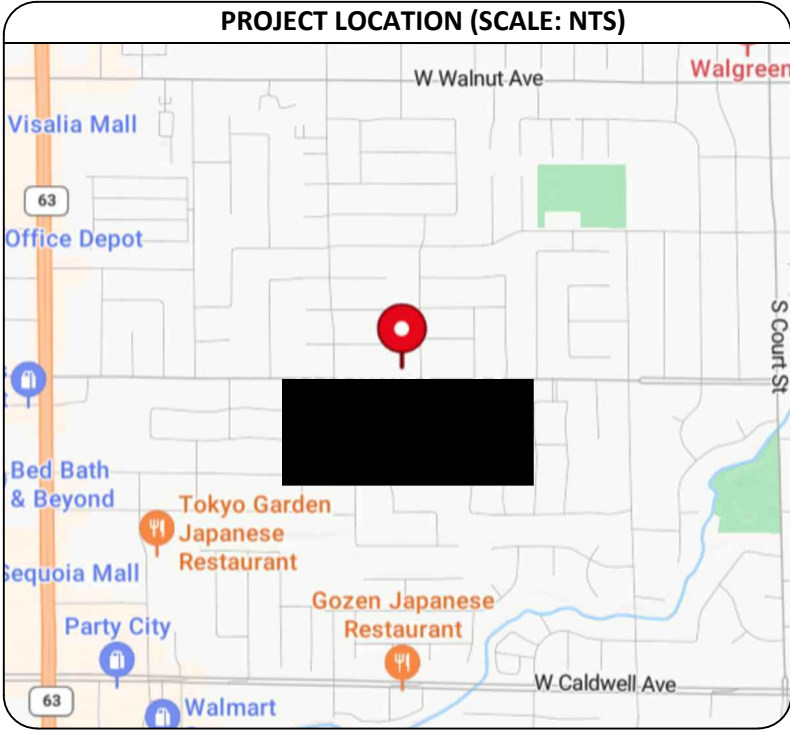
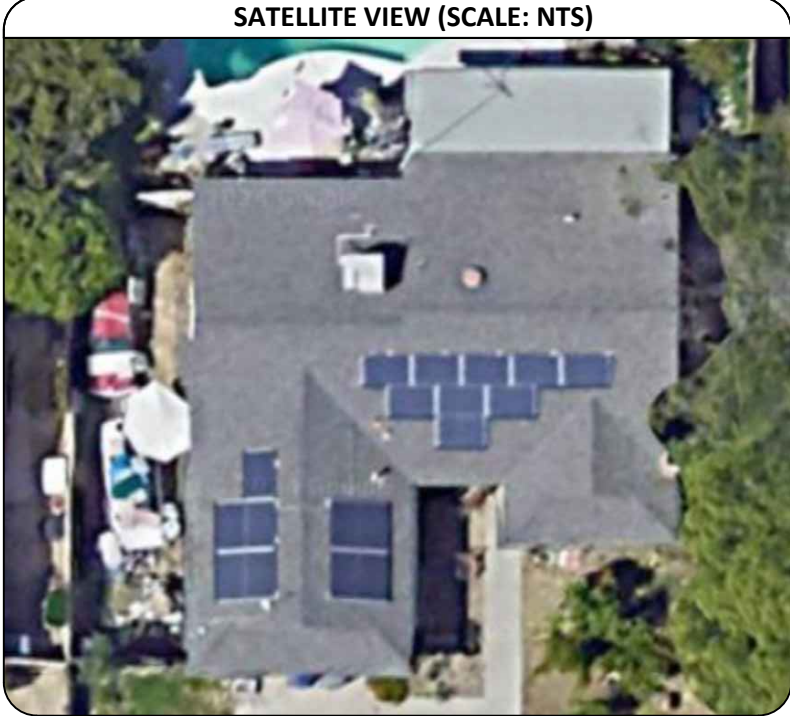


PROJECT: PV SYSTEM DETAILS	
[DC] SYSTEM SIZE	[AC] SYSTEM SIZE
(12) MODULES x 400W	(1) INVERTER x 3800W
4.800 kW	3.800 kW



PROJECT: EXISTING PV SYSTEM DETAILS	
[DC] SYSTEM SIZE	[AC] SYSTEM SIZE
(18) MODULES x 275W	(01) INVERTER x 3800W
4.950 kW	3.800 kW

- SCOPE OF WORK
- PROPOSED PROJECT TO INSTALL GRID- TIED PV + BATTERY SYSTEM ON AN EXISTING ROOFTOP STRUCTURE AT A RESIDENTIAL LOCATION: 1234 STREET NAME, CITY, CA 9874. COORDINATES: 36.305969, -119.303095
 - THE PV MODULES WILL BE SECURED ON ROOF USING PRE-ENGINEERED ROOF RACKING SYSTEM.
 - INSTALLED PV SYSTEM WILL BE THEN CONCECTED TO HOMEOWNER'S SERVICE EQUIPMENT USING APPLICABLE INTERCONCECTION METHOD (AS PER AHJ).
 - THE ENERGY PRODUCED BY PV SYSTEM WILL BE USED TO POWER THE HOME APPLIANCES. EXCESS POWER IS SENT BACK TO THE UTILITY GRID.

- APPLICABLE GOVERNING CODES & NOTES
- THIS PROJECT SHALL COMPLY WITH THE FOLLOWING CODES:
- 2022 CALIFORNIA BUILDING CODE
 - 2022 CALIFORNIA RESIDENTIAL CODE
 - 2022 CALIFORNIA MECHANICAL CODE
 - 2022 CALIFORNIA PLUMBING CODE
 - 2022 CALIFORNIA ENERGY CODE
 - 2022 CALIFORNIA FIRE CODE
 - 2022 CALIFORNIA ELECTRICAL CODE
- ALL OTHER RELATED STATE, LOCAL & AHJ AMENDMENTS.

- DESIGN CRITERIA
- CONSTRUCTION: SINGLE FAMILY DWELLING (SFD)
 - NO. OF STORIES: SINGLE STORY
 - EXPOSURE CATEGORY : B, RISK CATEGORY : II
 - WIND SPEED (ASCE 7-16) : 94 MPH
 - GROUND SNOW LOAD (ASCE 7-16) : 0 lb/sqft
 - ASHRAE EXTREME LOW: -3°C
 - ASHRAE HIGH TEMP (2% AVG.) : 37°C
 - ASHRAE DISTANCE ABOVE ROOF (7/8") : 59°C

SHEET INDEX	
PV-0	PROJECT SUMMARY
PV-1	SITE PLAN
PV-2	ROOF ATTACHMENT DETAILS
PV-3	BOM & STRING LAYOUT
S-1	STRUCTURAL DETAILED DIAGRAM
S-2	EQUIPMENT ELEVATION DIAGRAM
E-1 & E-2	ELECTRICAL DIAGRAM & CALCULATION
E-3	PLACARDS
E.S	EQUIPMENT SPECIFICATIONS

PROJECT INFORMATION	
(#) PV MODULES	(12) QCELLS Q.PEAK DUO BLK ML-G10+/t 400
(#)OPTIMIZERS	(12) SOLAREEDGE POWER OPTIMIZER S440
(#)INVERTER	(1) SOLAREEDGE ENERGY HUB SE3800H-US INVERTER
(#)BATTERY	(01) SOLAREEDGE ENERGY BANK BATTERY (10KW) (BAT-10K1PS0B-02)
SYSTEM OUTPUT CURRENT	16.00A
STRING	1 STRING OF 12 MODULE
ROOF RACKING SYSTEM	K2 SPLICE FOOT X W/ K2 CROSSRAIL 44-X
POINT OF INTERCONNECTION	100A LOAD SIDE BREAKER IN EXTERIOR METER-MAIN SERVICE PANEL WITH 100A MAIN BREAKER, 125A BUSBAR, 120/240V, 3W, 1φ
UTILITY	SCE
AHJ	VISALIA CITY

- GENERAL NOTES
- ALL COMPONENTS ARE UL LISTED & CEC CERTIFIED, WHERE WARRANTED
 - A DISCONNECTING MEANS SHALL BE INSTALLED AT A READILY ACCESSIBLE LOCATION EITHER OUTSIDE OF THE BUILDING OR STRUCTURE OR INSIDE NEAREST POINT OF ENTRANCE OF THE SYSTEM CONDUCTORS.
 - HEIGHT OF THE INTEGRATED AC/DC DISCONNECT SHALL NOT EXCEED 6'-7" PER CEC CODE 240.24.
 - A GROUNDING ELECTRODE SYSTEM IN ACCORDANCE WITH CEC 690.47 AND 250.50 THROUGH 60 AND 250-166 SHALL BE PROVIDED. PER CEC
 - GROUNDING ELECTRODE SYSTEM OF EXISTING BUILDING MAY BE USED AND BONDED TO THE SERVICE ENTRANCE. IF EXISTING SYSTEM IS INACCESSIBLE OR INADEQUATE A SUPPLEMENTAL GROUNDING ELECTRODE WILL BE USED AT THE INVERTER LOCATION CONSISTING OF A UL LISTED 8 FT. GROUND ROD WITH ACORN CLAMP. GROUNDING ELECTRODE CONDUCTORS SHALL BE NO LESS THAN #8 AWG AND NO LARGER THAN #6 AWG COPPER AND BONDED TO THE EXISTING GROUNDING ELECTRODE TO PROVIDE FOR A COMPLETE SYSTEM.
 - PHOTOVOLTAIC MODULES ARE TO BE CONSIDERED NON-COMBUSTIBLE.
 - PHOTOVOLTAIC INSTALLATION WILL NOT OBSTRUCT ANY PLUMBING, MECHANICAL, OR BUILDING ROOF VENTS.
 - WATER PROOF CONCECTORS AND HUBS SHALL BE USED WHERE APPLICABLE PER CEC 312.2 AND 314.15.
 - GROUNDING/BONDING BUSHINGS SHALL BE INSTALLED WHERE APPLICABLE PER CEC 250.92.
 - ALL EXTERIOR RACEWAYS ON WALLS SHOULD BE EMT UNLESS NOTED OTHERWISE.
 - CONTRACTOR SHALL INSTALL EQUIPMENT IN ACCORDANCE WITH MANUFACTURER INSTRUCTIONS & NOTHING IN THIS DOCUMENT SHALL BE INTERPRETED IN A WAY THAT OVERRIDES THEM.
 - CONTRACTOR IS RESPONSIBLE FOR VERIFICATION OF ALL CONDITIONS, DIMENSIONS & DETAILS IN THIS DOCUMENT. ALL DIMENSIONS SHALL BE VERIFIED BY SUBCONTRACTOR BEFORE INSTALLATION.
- PHOTOVOLTAIC SYSTEM FIRE CLASSIFICATION LISTING IN ACCORDANCE WITH UL1703 STANDARD.

- ELECTRICAL NOTES
- THE EQUIPMENT AND ALL ASSOCIATED WIRING AND INTERCONCECTION SHALL BE INSTALLED ONLY BY QUALIFIED PEOPLE. A QUALIFIED PERSON IS ONE WHO HAS SKILLS AND KNOWLEDGE RELATED TO THE CONSTRUCTION AND OPERATION OF THE ELECTRICAL EQUIPMENT AND INSTALLATIONS AND HAS RECEIVED SAFETY TRAINING TO RECOGNIZE AND AVOID THE HAZARDS INVOLVED. (CEC690.4(E) AND 705.6)
 - LOCAL UTILITY PROVIDER SHALL BE NOTIFIED PRIOR TO USE AND ACTIVATION OF ANY SOLAR PHOTOVOLTAIC INSTALLATION. FOR A LINE SIDE TAP CONCECTION, UTILITY NEEDS TO BE NOTIFIED WELL IN ADVANCE TO COORDINATE BUILDING ELECTRICAL SHUT OFF.
 - NEW CONDUIT ROUTING SHOWN IS ESSENTIALLY SCHEMATIC. SUBCONTRACTOR SHALL LAY OUT RUNS TO SUIT FIELD CONDITIONS AND THE COORDINATION REQUIREMENTS OF OTHER TRADES.
 - ARRAY WIRING SHOULD NOT BE READILY ACCESSIBLE EXCEPT TO QUALIFIED PERSONNEL.
 - ALL EXTERIOR CONDUIT, FITTINGS, AND BOXES SHALL BE WATERTIGHT AND APPROVED FOR USE IN WET LOCATIONS .
 - WIRING METHODS FOR PV SYSTEM CONDUCTORS AREN'T PERMITTED WITHIN 10 IN. OF THE ROOF DECKING OR SHEATHING EXCEPT WHERE LOCATED DIRECTLY BELOW THE ROOF SURFACE THAT'S COVERED BY PV MODULES AND ASSOCIATED EQUIPMENT WIRING
 - BACK-FED BREAKER MUST BE AT THE OPPOSITE END OF BUS BAR FROM THE MAIN BREAKER OR MAIN LUG SUPPLYING CURRENT FROM THE UTILITIES.
 - ALL CONDUCTORS AND WIRE TIES EXPOSED TO SUNLIGHT ARE LISTED AS UV RESISTANT.
 - CONTRACTOR SHALL FOLLOW ALL ELECTRICAL EQUIPMENT LABELING REQUIREMENTS ACCORDING TO CEC.
 - MEASURE THE LINE-TO-LINE AND LINE-TO-NEUTRAL VOLTAGE OF ALL SERVICE ENTRANCE CONDUCTORS PRIOR TO INSTALLING ANY SOLAR EQUIPMENT. THE VOLTAGES FOR THE 240VAC RATED.
 - ALL EXTERIOR CONDUIT TO BE PAINT MATCHED TO THE HOMES COLORS

DATE	06/07/2024
PROJECT ID	OPS 034-598
CREATED BY	MSB
CHECKED BY	SSN
SCALE	NTS
SHEET SIZE	11"X 17"
	ANSI B

CLIENT RESIDENCE
1234 STREET NAME, CITY, CA 9874
APN NUMBER: ----

HOMEOWNER INFO

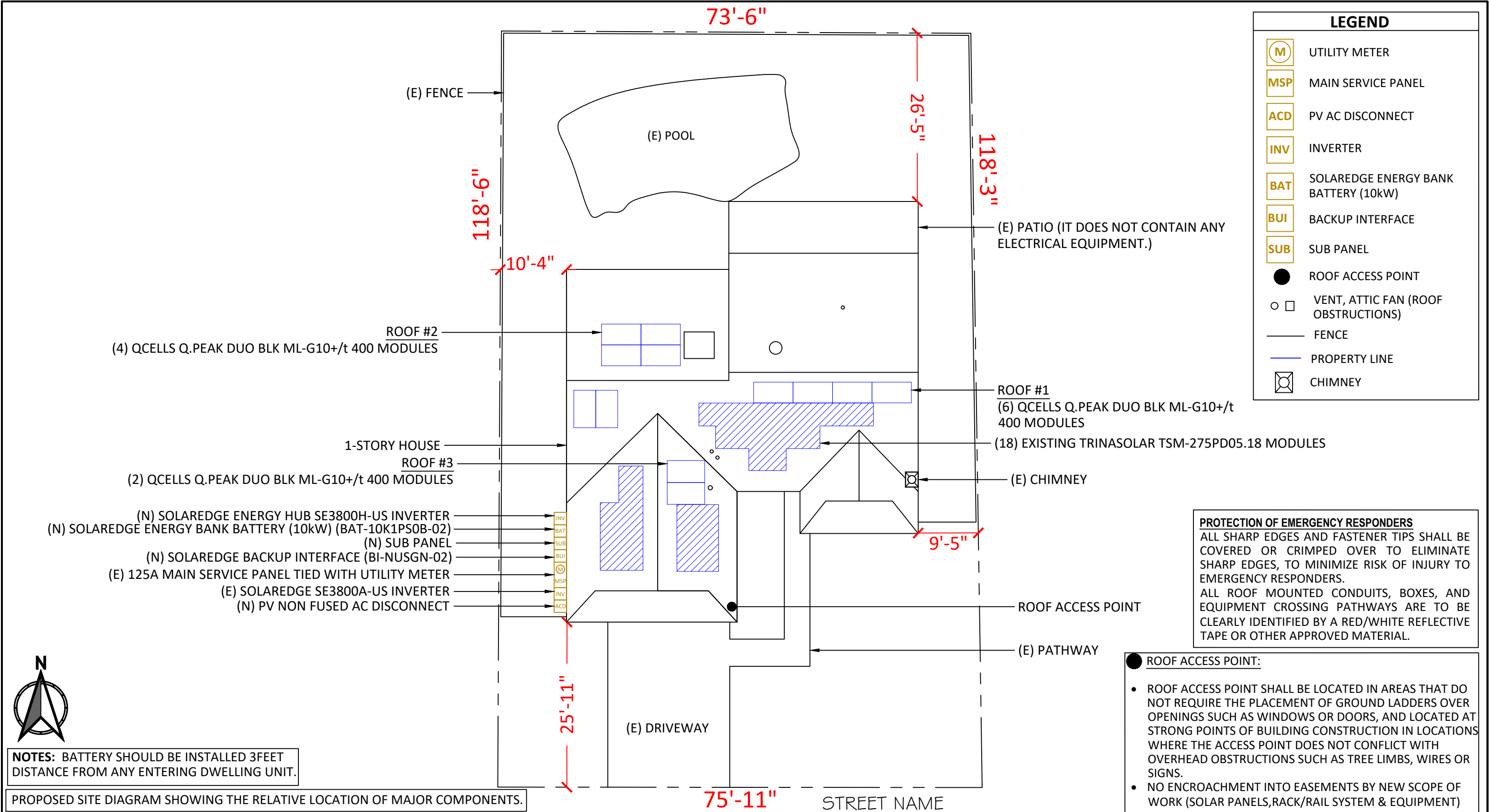
#	REVISION/UPDATE	DATE

SIGNED & STAMP BY

PV-0
SHEET NO.

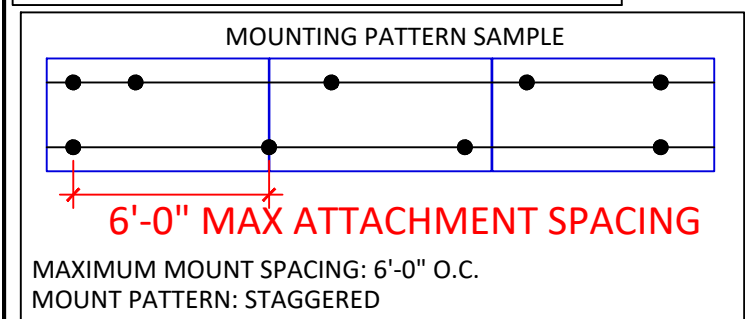
PROJECT SUMMARY

SHEET INFORMATION

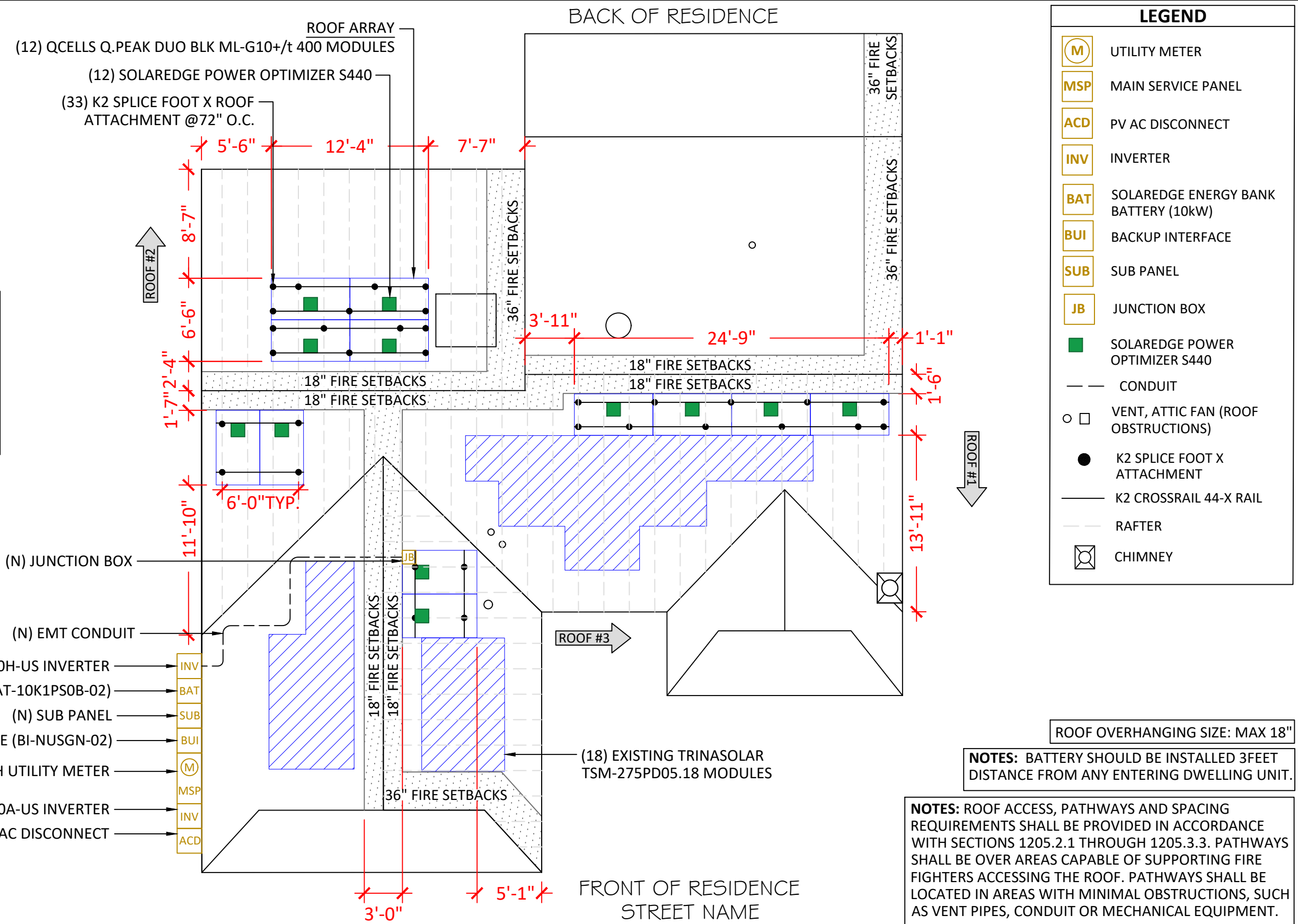


	DATE	06/07/2024	CLIENT RESIDENCE 1234 STREET NAME, CITY, CA 9874 APN NUMBER: ----	#	REVISION/UPDATE	DATE	SIGNED & STAMP BY	PV-1 SHEET NO. SITE PLAN
	PROJECT ID	OPS 034-598						
	CREATED BY	MSB						
	CHECKED BY	SSN						
	SCALE	1/14" = 1'-0"						
	SHEET SIZE	11"X 17"						
	ANSI B	HOMEOWNER INFO						SHEET INFORMATION

ROOF DETAILS			
MOUNTING PLANE	ROOF #1	ROOF #2	ROOF #3
NO. OF MODULES	6	4	2
AZIMUTH	180°	360°	90°
ROOF TILT	18°	18°	18°
ROOF TYPE	COMP. SHINGLE	COMP. SHINGLE	COMP. SHINGLE
RAFTER SIZE	2" X 4"	2" X 4"	2" X 4"
RAFTER SPACING	24" O.C	24" O.C	24" O.C
NEW ARRAY AREA	253.45		
EXISTING ARRAY AREA	337.05		
(EXISTING + NEW ARRAY AREA/ TOTAL ROOF AREA) X 100% = (590.50/2901.67) X 100% = 20.35%			



NOTES: ATTACHMENT POINTS ARE NOT EXACT. CONTRACTOR MAY NEED TO ADJUST MOUNT LOCATION. EXISTING ROOF CONDITION & RAFTER LOCATIONS ARE SUBJECT TO FIELD VERIFICATION. ACTUAL LOCATION MAY DIFFER. INSTALL PER MANUFACTURER SPECIFICATIONS. SEE SUPPLEMENTAL DOCUMENTS FOR ATTACHMENT & STRUCTURAL DOCUMENTATION.



DATE	06/07/2024
PROJECT ID	OPS 034-598
CREATED BY	MSB
CHECKED BY	SSN
SCALE	1/8" = 1'-0"
SHEET SIZE	11"X 17"
	ANSI B

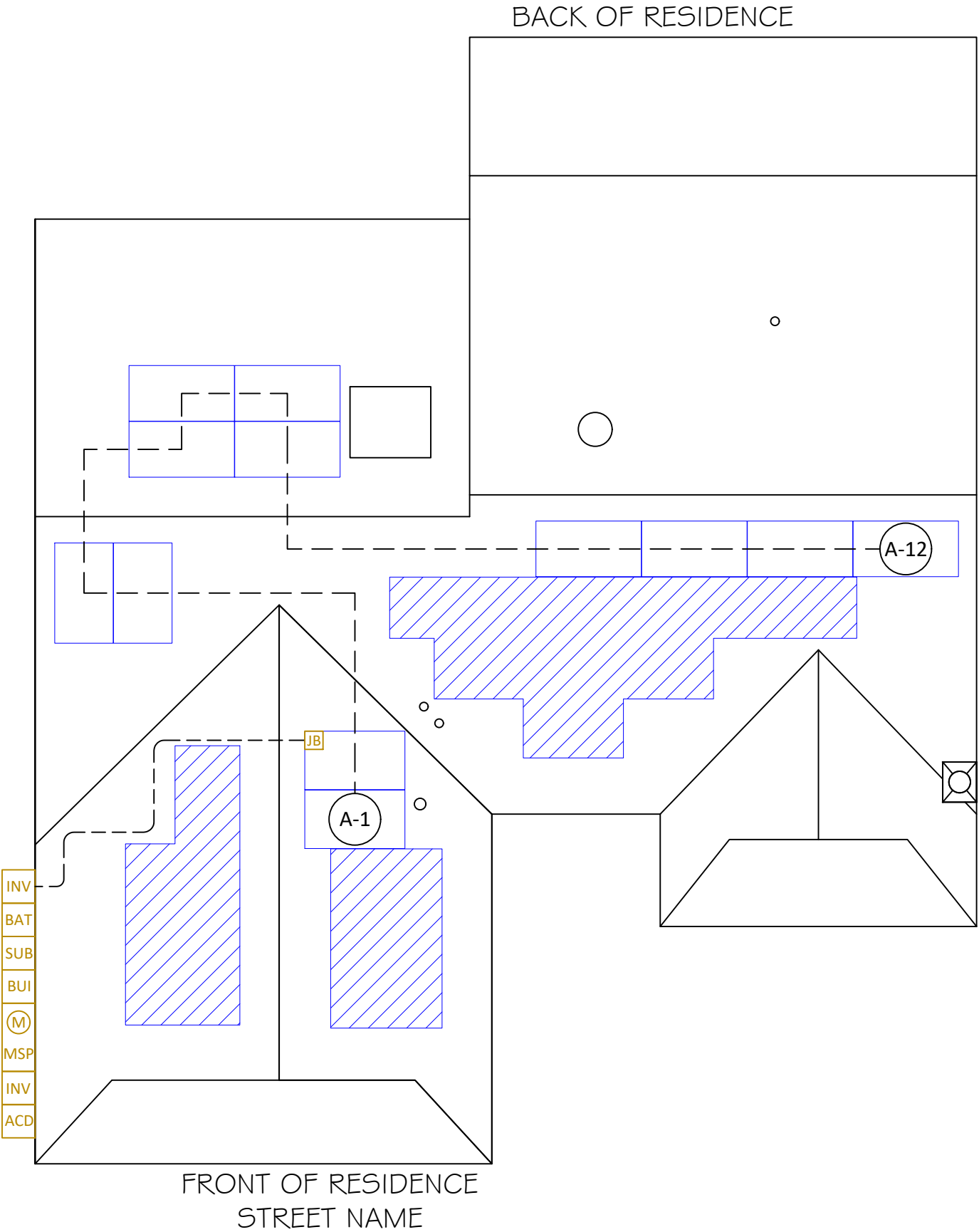
CLIENT RESIDENCE
1234 STREET NAME, CITY, CA 9874
APN NUMBER: ----

[illegible]

SIGNED & STAMP BY

BILL OF MATERIALS		
EQUIPMENT	QTY	DESCRIPTION
SOLAR PV MODULE	12	QCELLS Q.PEAK DUO BLK ML-G10+/t 400 MODULES
OPTIMIZER	12	SOLAREEDGE POWER OPTIMIZER S440
INVERTER	1	SOLAREEDGE ENERGY HUB SE3800H-US INVERTER
JUNCTION BOX	1	JUNCTION BOX, 600V, NEMA 3R, UL LISTED
BATTERY	1	SOLAREEDGE ENERGY BANK BATTERY (10kW) (BAT-10K1PS0B-02)
PV AC DISCONNECT	1	30A NON-FUSED PV AC DISCONNECT, 120/240V 2P, NEMA 3R
BACKUP INTERFACE	1	SOLAREEDGE BACKUP INTERFACE (BI-NUSGN-02)
SUB PANEL	1	SUB PANEL, 100A RATED, MLO, NEMA 3R, 240V
ATTACHMENT	33	K2 SPLICE FOOT X
ATTACHMENT	132	M5 X 60 LAG SCREWS
ATTACHMENT	33	K2 SOLAR SEAL BUTYL PAD
ATTACHMENT	33	T-BOLTS & HEX NUT SET
RAIL	10	K2 CROSSRAIL 44-X RAIL (166")
BONDED SPLICE	2	BONDED SPLICE
CLAMPS	14	CROSSRAIL MID CLAMP
CLAMPS	20	CROSSRAIL END CLAMP
GROUNDING LUG	5	GROUNDING LUG

ARRAY WEIGHT (LOAD CALC'S)		
Number of Modules	12	
Module Weight	48.5	LBS
Total Module (Array) Weight	582.00	LBS
Number of Attachment point	33	
Mounting System Weight (Per Module)	0.5	LBS
Mounting System Weight	16.50	LBS
Total System Weight (Module Weight + Mounting System Weight)	598.50	LBS
Weight at Each Attachment Point (Array Weight / Number of Attachment Point)	17.64	LBS
Module Area (74"x41.1")	21.12	SqFt
New Total Array Area	253.45	SqFt
Existing Total Array Area	337.05	SqFt
Total Array Area	590.50	SqFt
Distributed Load (Total System Weight / Total Array Area)	2.36	Per SqFt
Total Roof Area	2901.67	SqFt
Total Percentage of Roof Covered Total Array Area / Total Roof Area)*100	20.35%	



LEGEND	
	UTILITY METER
	MAIN SERVICE PANEL
	PV AC DISCONNECT
	INVERTER
	SOLAREEDGE ENERGY BANK BATTERY (10kW)
	BACKUP INTERFACE
	SUB PANEL
	JUNCTION BOX
	EMT CONDUIT
	VENT, ATTIC FAN (ROOF OBSTRUCTIONS)
	CHIMNEY
STRING LEGEND	
	STRING #1 - 12 MODULES

NOTES: BATTERY SHOULD BE INSTALLED 3FEET DISTANCE FROM ANY ENTERING DWELLING UNIT.

--

DATE	06/07/2024
PROJECT ID	OPS 034-598
CREATED BY	MSB
CHECKED BY	SSN
SCALE	1/8" = 1'-0"
SHEET SIZE	11"X 17"
	ANSI B

CLIENT RESIDENCE 1234 STREET NAME, CITY, CA 9874 APN NUMBER: ----	
HOMEOWNER INFO	

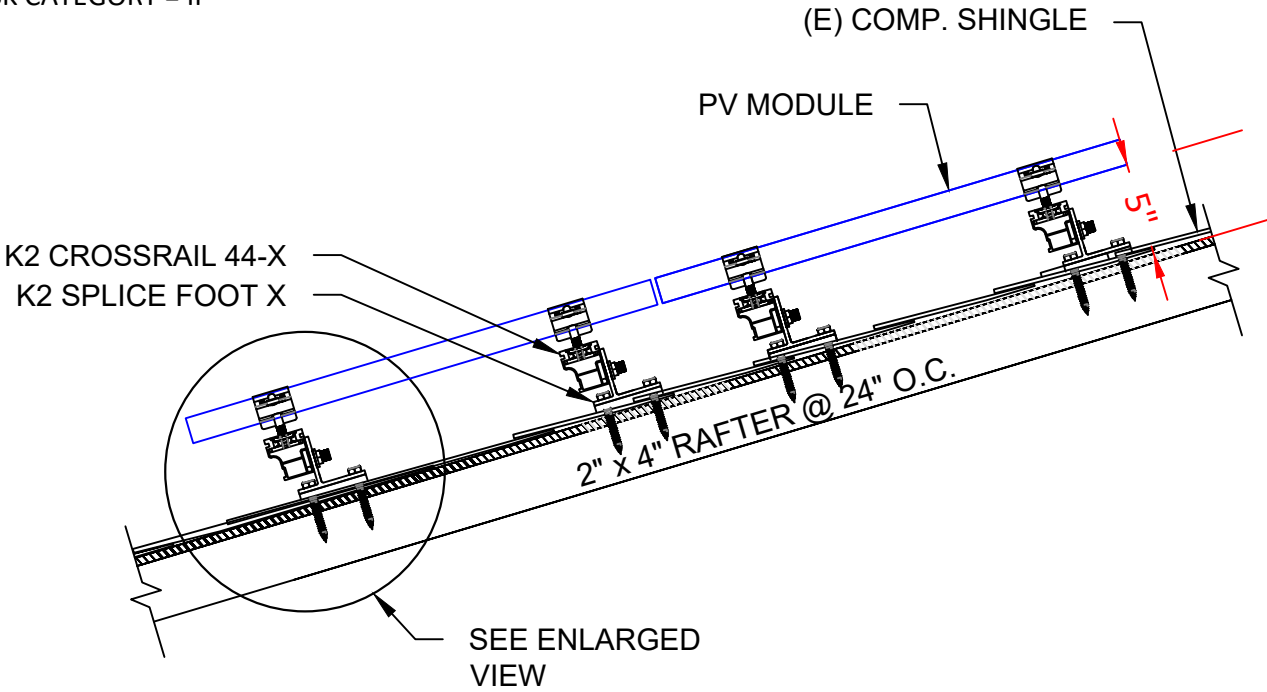
#	REVISION/UPDATE	DATE

SIGNED & STAMP BY

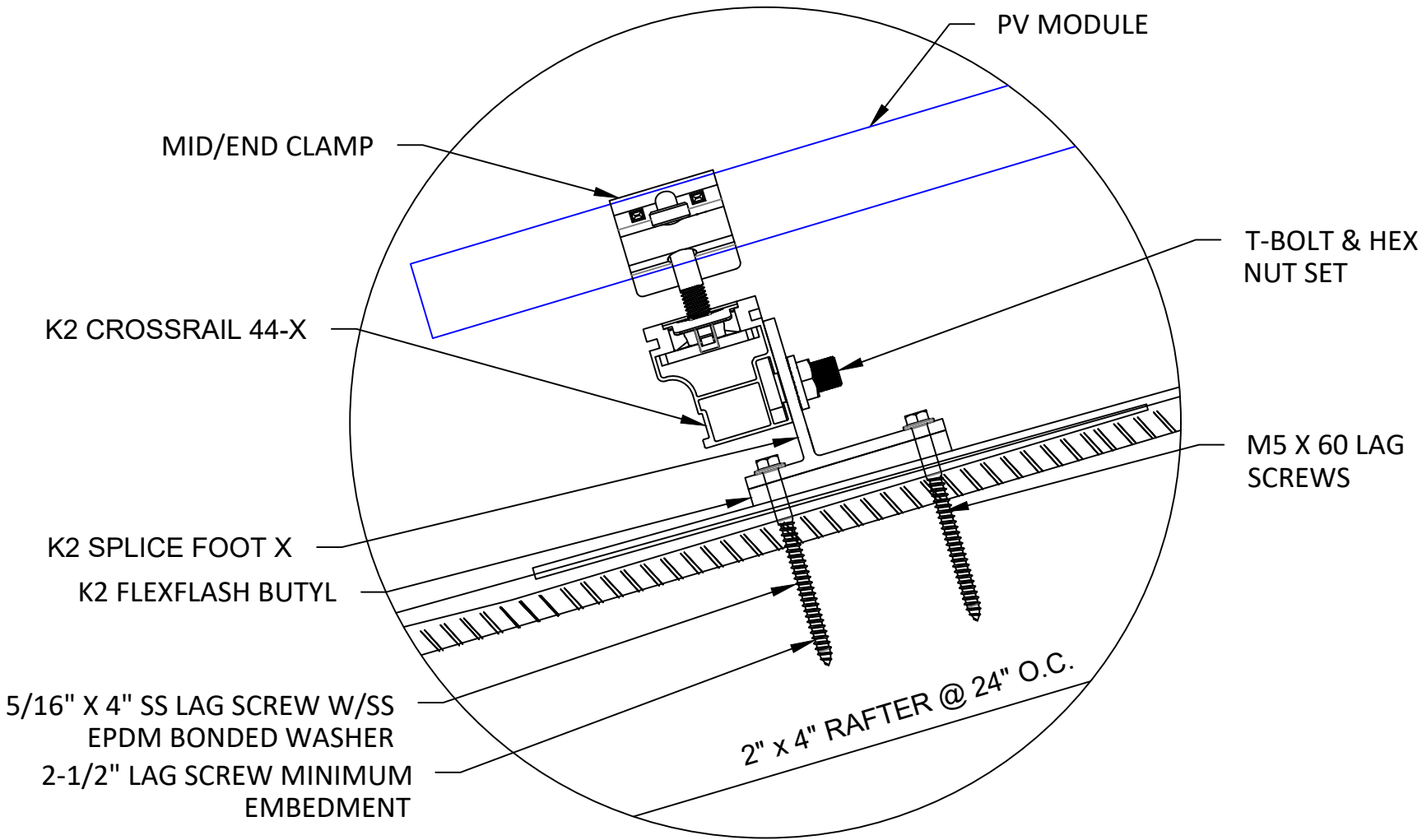
PV-3 SHEET NO. BOM & STRING LAYOUT SHEET INFORMATION
--

STRUCTURAL NOTES:

- 1. THE SOLAR PANELS ARE TO BE MOUNTED TO THE ROOF FRAMING USING THE K2 CROSSRAIL 44-X SYSTEM WITH K2 SPLICE FOOT X ATTACHMENTS.
- 2. THE MOUNTING FEET ARE TO BE SPACED AS SHOWN IN THE DETAILS, AND MUST BE STAGGERED TO ADJACENT FRAMING MEMBERS TO SPREAD OUT THE ADDITIONAL LOAD.
- 3. THE PROPOSED PV SYSTEM ADDS 2.36 PSF TO THE ROOF FRAMING SYSTEM.
- 4. ROOF LIVE LOAD = 20 PSF
- 5. GROUND SNOW LOAD = 0 lb/sqft
- 6. WIND SPEED = 94 MPH
- 7. EXPOSURE CATEGORY = B
- 8. RISK CATEGORY = II



STRUCTURAL ELEVATION DIAGRAM



ATTACHMENT DETAILED DIAGRAM

NOTES: FLASHING SHALL BE APPLIED IN ACCORDANCE WITH MANUFACTURERS PRINTED INSTRUCTIONS. IF THERE IS ANY CONFLICT BETWEEN WHAT IS DEPICTED & INSTRUCTION AS PER MANUFACTURER, MANUFACTURERS INSTRUCTIONS SHALL SUPERSEDE.

DATE	06/07/2024
PROJECT ID	OPS 034-598
CREATED BY	MSB
CHECKED BY	SSN
SCALE	NTS
SHEET SIZE	11"X 17"
	ANSI B

CLIENT RESIDENCE

1234 STREET NAME, CITY, CA 9874

APN NUMBER: ----

HOMEOWNER INFO

#	REVISION/UPDATE	DATE

SIGNED & STAMP BY

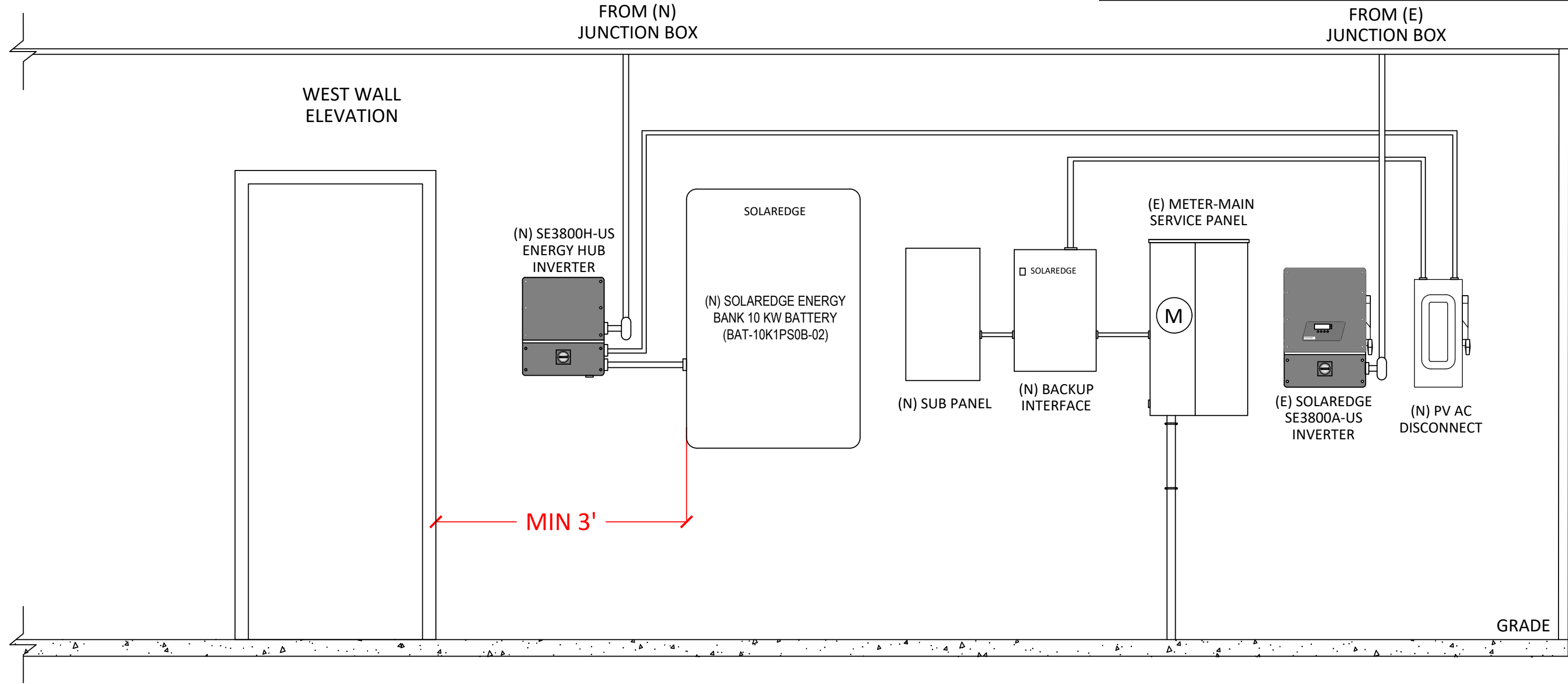
S-1

SHEET NO.

STRUCTURAL DETAILED DIAGRAM

SHEET INFORMATION

NOTES:
ESS UNIT LOCATION(S) AND DISTANCES RELATIVE TO ONE ANOTHER AND TO ALL RELEVANT BUILDING FEATURES AND OTHER EQUIPMENT, INCLUDING BUT NOT LIMITED TO THE FOLLOWING: (2020 LACFC §1206.4.2.1, §1206.4.3.1)
I.) INDIVIDUAL ESS UNITS SHALL BE SEPARATED FROM ONE ANOTHER BY 3 FEET.
II.) WHERE AN ESS IS INSTALLED ON AN EXTERIOR WALL OR OUTDOOR, ESS UNITS SHALL BE SEPARATED FROM ALL WINDOWS, DOORS, OPERABLE OPENINGS AND HVAC INLETS BY 3 FEET (IN ANY DIRECTION).



EQUIPMENT ELEVATION DIAGRAM

--	--

DATE	06/07/2024
PROJECT ID	OPS 034-598
CREATED BY	MSB
CHECKED BY	SSN
SCALE	NTS
SHEET SIZE	11"X 17"
	ANSI B

CLIENT RESIDENCE 1234 STREET NAME, CITY, CA 9874 APN NUMBER: ----	
HOMEOWNER INFO	

#	REVISION/UPDATE	DATE

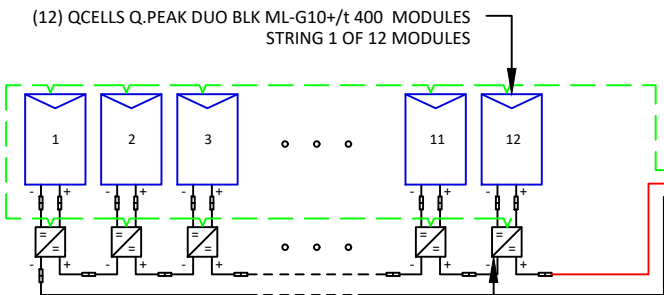
SIGNED & STAMP BY	

S-2	
SHEET NO.	
EQUIPMENT ELEVATION DIAGRAM	
SHEET INFORMATION	

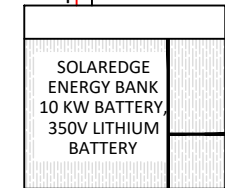
NEW PV SYSTEM DETAILS	
[DC] SYSTEM SIZE	[AC] SYSTEM SIZE
(12) MODULES x 400W	(1) INVERTER x 3800W
4.800 kW	3.800 kW

NEW SOLAR MODULE SPECIFICATIONS	
MANUFACTURER / MODEL	QCELLS Q.PEAK DUO BLK ML-G10+/t 400
VMP	38.09V
IMP	10.50A
VOC	45.55V
ISC	11.07A
TEMP. COEFF. VOC	-0.27 %/K
MODULE DIMENSION	74" (L) x 41.1" (W)
PANEL WATTAGE	400W

EXISTING SOLAR MODULE SPECIFICATIONS	
MANUFACTURER / MODEL	TRINASOLAR TTSM-275PD05.18
VMP	31.1V
IMP	8.84A
VOC	38.1V
ISC	9.32A
TEMP. COEFF. VOC	-0.32 %/°C
MODULE DIMENSION	65.0" (L) x 39.1" (W)
PANEL WATTAGE	275W



SolarEdge Power Optimizer S440 Rated
DC Input Power - 440 watts
Maximum Input Voltage - 60 Vdc
MPPT Range - 8 to 60 Vdc
Maximum Input Current - 14.5 Adc
Maximum Output Current - 15 Adc String
Limitations - 8 to 25 Optimizers,
5700 watts STC per string maximum



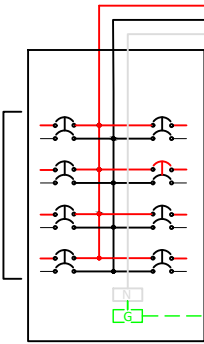
(N) SOLAREEDGE ENERGY BANK
10 KW BATTERY

(N) JUNCTION BOX:
600V, NEMA 3R,
UL LISTED
(READILY ACCESSIBLE)

(N) SOLAREEDGE ENERGY HUB SE3800H-US
INVERTER 99% CEC EFFICIENCY @ 240VAC
3800WAC CONTINUOUS MAXIMUM OUTPUT
CURRENT 16A GROUND-FAULT PROTECTION
PROVIDED PER NEC ARTICLE 690.35

RELOCATED SOME
EXISTING LOADS

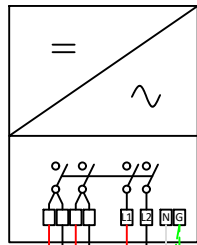
(N) PV AC DISCONNECT, 30A
NON-FUSED, LOCKABLE,
120/240V 2P, NEMA 3R,
UL-LISTED



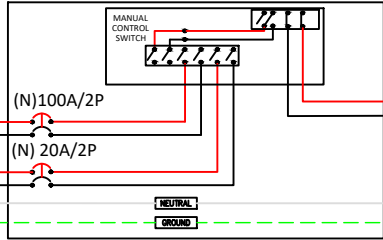
(N) SUB PANEL, 100A
RATED, MLO, 1-PH, 3-W,
240V

(E) SOLAREEDGE SE3800A-US
MAXIMUM OUTPUT CURRENT 16A
GROUND-FAULT PROTECTION
PROVIDED PER NEC ARTICLE 690.35

(E) JUNCTION BOX
600V, NEMA 3R,
UL LISTED

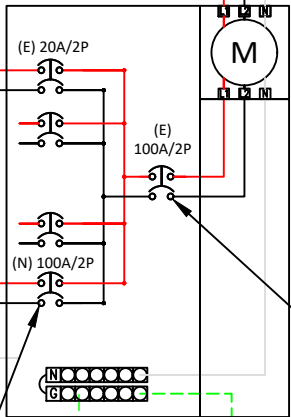


(N) SOLAREEDGE BI-NUSGN-02 BACKUP
INTERFACE, 240, NEMA 3R, UL LISTED



POINT OF INTERCONNECTION,
100A LOAD SIDE BREAKER

TO UTILITY GRID



BI-DIRECTIONAL
UTILITY METER #222014-001014
1-PH, 3-W, 120V/240V
UTILITY COMPANY - SCE

(E) EXTERIOR METER-MAIN
SERVICE PANEL, 125A
RATED, 1-PH, 3-W, 240V

(E) 100A/2P MAIN BREAKER
TO HOUSE, 240V

EXISTING GROUNDING
ELECTRODE CONDUCTOR

NOTES: BATTERY SHOULD BE INSTALLED 3FEET
DISTANCE FROM ANY ENTERING DWELLING UNIT.

EXISTING PV SYSTEM DETAILS	
[DC] SYSTEM SIZE	[AC] SYSTEM SIZE
(18) MODULES x 275W	(1) INVERTER x 3800W
4.950 kW	3.800 kW

THE SUMMATION RULE - NEC - 705.12(B)(3)(3)

NOTES: TOTAL RATING OF ALL OVER CURRENT DEVICES
EXCLUDING MAIN SUPPLY OVER CURRENT DEVICE SHALL
NOT EXCEED AMPACITY OF BASBUR 705.12(B)(3)(3).

INVERTER SPECIFICATIONS	
MANUFACTURER / MODEL	SOLAREEDGE ENERGY HUB SE3800H-US
NOMINAL OUTPUT VOLTAGE	240VAC
CONTINUOUS OUTPUT CURRENT	16A

DATE	06/07/2024
PROJECT ID	OPS 034-598
CREATED BY	MSB
CHECKED BY	SSN
SCALE	NTS
SHEET SIZE	11"X 17"
	ANSI B

HOMEOWNER INFO

CLIENT RESIDENCE
1234 STREET NAME, CITY, CA 9874
APN NUMBER: ----

#	REVISION/UPDATE	DATE

SIGNED & STAMP BY

E-1
SHEET NO.
ELECTRICAL
DIAGRAM &
CALCULATIONS
SHEET INFORMATION

EQUIPMENT DETAILS	
(#) PV MODULES	(12) QCELLS Q.PEAK DUO BLK ML-G10+/t 400
(#) INVERTER	(1) SOLAREEDGE ENERGY HUB SE3800H-US
(#) OPTIMIZERS	(12) SOLAREEDGE POWER OPTIMIZER S440
(#) BATTERY	(01) SOLAREEDGE ENERGY BANK BATTERY (10KW)(BAT-10K1PS0B-02)
SYSTEM OUTPUT CURRENT	16.00A
STRING	1 STRING OF 12 MODULE
POINT OF INTERCONNECTION	100A LOAD SIDE BREAKER IN EXTERIOR METER-MAIN SERVICE PANEL WITH 100A MAIN BREAKER, 125A BUSBAR, 120/240V, 3W, 1ϕ
UTILITY	SCE
AHJ	VISALIA CITY

AMBIENT TEMPERATURE SPECS	
RECORD LOW TEMP	-3°C
AMBIENT TEMP (HIGH TEMP 2%)	37°C
CONDUIT HEIGHT	7/8"
ROOF TOP TEMP	90°C
CONDUCTOR TEMPERATURE RATE	59°C
MODULE TEMPERATURE COEFFICIENT OF VOC	-0.27 %/K

BATTERY SPECIFICATION	
MODEL	SOLAREEDGE ENERGY BANK BATTERY (10KW)(BAT-10K1PS0B-02)
AMBIENT TEMPERATURE RANGE	-10°C TO 50°C
VOLTAGE RANGE	350-450 VDC
USABLE CAPACITY	9.7 KWH
INTERCONNECTION	SINGLE-PHASE

ID	CONDUCTOR			CONDUIT	# OF PARALLEL CIRCUITS	CURRENT-CARRYING CONDUCTORS IN CONDUIT	CONDUIT FILL PERCENT	OCPD	EGC		TEMP. CORR. FACTOR		CONDUIT FILL FACTOR	CONT. CURRENT (INVERTER CONTINUOUS CURRENT x 1.25)				MAX CURRENT	BASE AMP.	DERATED AMP.	LENGTH	VOLTAGE DROP
A	10 AWG	PV WIRE		OPEN AIR	1	2	N/A	N/A	6 AWG	BARE COPPER	0.71	(59°C)	N/A	15A	x	1.25	=	18.8A	N/A	N/A	88FT	0.03%
B	10 AWG	THWN-2	COPPER	MIN 0.75" Dia EMT	2	4	19.09%	N/A	8 AWG	THWN-2, COPPER	0.91	(37°C)	0.8	15A	x	1.25	=	18.8A	35A	35A x 0.8 x 0.91 = 25.5A	34FT	0.11%
C	10 AWG	THWN-2	COPPER	MIN 0.75" Dia EMT	1	2	11.45%	N/A	8 AWG	THWN-2, COPPER	0.91	(37°C)	1	14.3A	x	1.25	=	17.9A	35A	35A x 1 x 0.91 = 31.9A	5FT	0.08%
D	10 AWG	THWN-2	COPPER	MIN 0.75" Dia EMT	1	3	15.27%	N/A	8 AWG	THWN-2, COPPER	0.91	(37°C)	1	16.0A	x	1.25	=	20.0A	35A	35A x 1 x 0.91 = 31.9A	12FT	0.26%
E	10 AWG	THWN-2	COPPER	MIN 0.75" Dia EMT	1	3	15.27%	20A	8 AWG	THWN-2, COPPER	0.91	(37°C)	1	16.0A	x	1.25	=	20.0A	35A	35A x 1 x 0.91 = 31.9A	5FT	0.12%
F	2 AWG	THWN-2	COPPER	MIN 1.25" Dia EMT	1	3	30.01%	100A	8 AWG	THWN-2, COPPER	0.91	(37°C)	1	80A	x	1.25	=	100.0A	115A	115A x 1 x 0.91 = 104.7A	5FT	0.08%
G	2 AWG	THWN-2	COPPER	MIN 1.25" Dia EMT	1	3	30.01%	100A	8 AWG	THWN-2, COPPER	0.91	(37°C)	1	80A	x	1.25	=	100.0A	115A	115A x 1 x 0.91 = 104.7A	5FT	0.08%

DATE	06/07/2024
PROJECT ID	OPS 034-598
CREATED BY	MSB
CHECKED BY	SSN
SCALE	NTS
SHEET SIZE	11"X 17"
	ANSI B

CLIENT RESIDENCE

1234 STREET NAME, CITY, CA 9874

APN NUMBER: ----

HOMEOWNER INFO

#	REVISION/UPDATE	DATE

SIGNED & STAMP BY

E-2

SHEET NO.

ELECTRICAL
DIAGRAM &
CALCULATIONS

SHEET INFORMATION

1

! WARNING

ELECTRIC SHOCK HAZARD

DO NOT TOUCH TERMINALS.
TERMINALS ON BOTH LINE AND LOAD SIDES MAY
BE ENERGIZED IN THE OPEN POSITION

LABEL LOCATION:
MAIN SERVICE PANEL/AC DISCONNECT/INVERTER/
AC COMBINER
2022 CEC 690.13(B)

2

! WARNING

ELECTRIC SHOCK HAZARD

IF GROUND FAULT IS INDICATED
ALL NORMALLY GROUNDED
CONDUCTORS MAY BE
UNGROUNDED AND ENERGIZED

LABEL LOCATION:
INVERTER(S), ENPHASE ENVOY ENCLOSURE (IF
APPLICABLE).
PER CODE(S): CEC 2022: 690.5(C)

3

PHOTOVOLTAIC AC DISCONNECT

RATED AC OUTPUT CURRENT 16.00A

NOMINAL OPERATING AC VOLTAGE 240 VAC

LABEL LOCATION:
MAIN SERVICE PANEL/AC DISCONNECT
2022 CEC 690.53

4

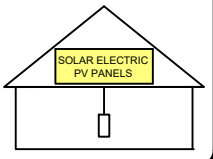
RAPID SHUTDOWN SWITCH
FOR SOLAR PV SYSTEM

LABEL LOCATION:
INVERTER
AT OR WITHIN 3' OF THE DC COMBINER
SWITCH
2022 CEC 690.56(C)(3)

5

SOLAR PV SYSTEM EQUIPPED
WITH RAPID SHUTDOWN

TURN RAPID SHUTDOWN
SWITCH TO THE
"OFF" POSITION TO
SHUT DOWN PV SYSTEM
AND REDUCE
SHOCK HAZARD
IN THE ARRAY



LABEL LOCATION:
MAIN SERVICE PANEL
IF MSD IS OUTSIDE PLACE IT THERE / IF
MSD IS INSIDE PLACE ON THE AC DISCONNECT
2022 CEC 690.56(C)(1)(a)

6

PHOTOVOLTAIC SYSTEM
UTILITY DISCONNECT SWITCH

LABEL LOCATION:
AC DISCONNECT
2022 CEC 690.54

7

PV SOLAR BREAKER

DO NOT RELOCATE THIS
OVERCURRENT DEVICE

LABEL LOCATION:
MAIN SERVICE PANEL
2022 CEC 705.12(B)(2)(3)(c)

8

MAXIMUM VOLTAGE: 480 VDC
MAXIMUM CIRCUIT CURRENT: 15 ADC
MAX. RATED OUTPUT CURRENT OF THE
CHARGE CONTROLLER OR
DC-TO-DC-CONVERTER (IF
INSTALLED) 15 ADC

LABEL LOCATION:
INVERTER
PER 2022 CEC 690.53

9

BATTERY DISCONNECT

DO NOT RELOCATE
THIS OVERCURRENT
DEVICE

REQ'D BY: CEC 705.12(B)(2)(3)(b)
LABEL LOCATION:
BATTERY BREAKERS

10

CAUTION - TRI POWER SOURCE

FIRST SOURCE IS UTILITY ELECTRICAL GRID
SECOND SOURCE IS AC BATTERY
THIRD SOURCE IS PV INVERTER

LABEL LOCATION:
SUB PANEL OR BACKUP LOADS PANEL, BATTERY, INVERTER OUTPUT
PANEL, MAIN SERVICE PANEL, METER/MAIN AND PV AC DISCONNECT
2022 CEC 705.10

11

SOLAREDGE

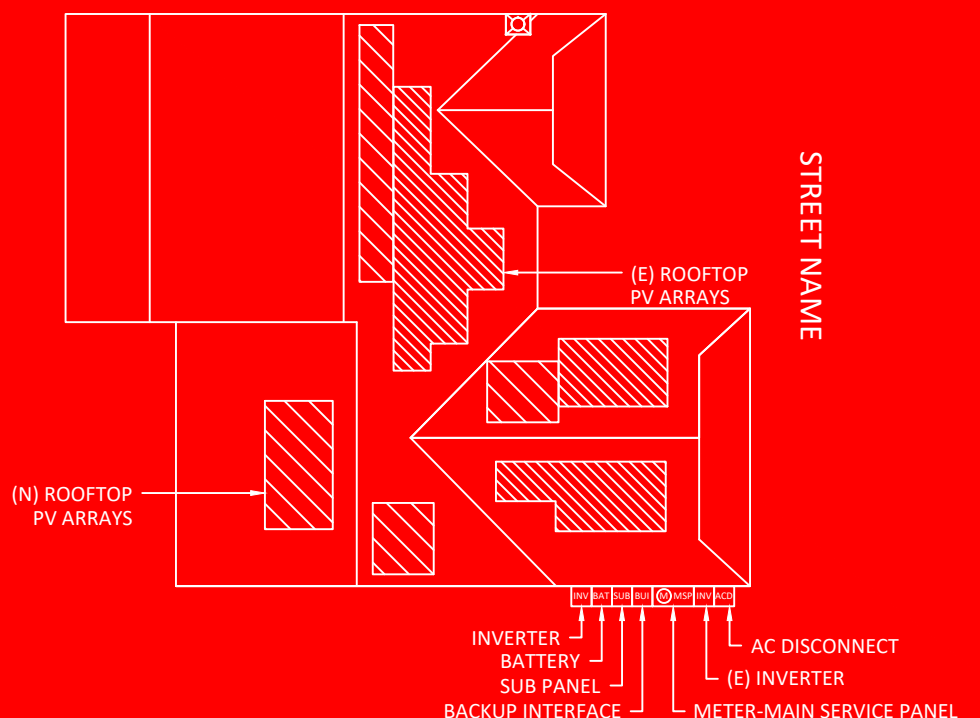
BAT-10K1P

NOMINAL ENERGY	9.7 KWH
RATED CAPACITY	94 AH
OUTPUT VOLTAGE (MIN-NOM-MAX)	350-400-450 VDC
CONTINUOUS OUTPUT POWER	5.0 KWDC
MAX CONTINUOUS OUTPUT CURRENT	14.3 ADC
PEAK OUTPUT POWER IN	
BACKUP / DURATION	7.5/10 KWDC/SEC
MAX SHORT CIRCUIT	
CURRENT / DURATION	1K/10 ADC/MSEC
ENCLOSURE	NEMA 3R / IPSS
OPERATING TEMPERATURE RANGE	-10°C to +50°C
STORAGE TEMPERATURE RANGE	-30°C to +60°C
BATTERY DESIGNATION	ICP/46/174/134/[1P305]E/-10+50/95

LABEL LOCATION:
BATTERY

CAUTION!

POWER TO THIS BUILDING IS ALSO SUPPLIED FROM THE
FOLLOWING SOURCES WITH DISCONNECTS LOCATED AS SHOWN.



LABEL LOCATION:
EACH SERVICE EQUIPMENT LOCATION AND AT THE LOCATION(S) OF THE SYSTEM DISCONNECT(S) FOR ALL ELECTRIC POWER
PRODUCTION SOURCES CAPABLE OF BEING INTERCONNECTED

ADHESIVE FASTENED SIGNS:

- LABELS SHALL BE MADE OF RED PLASTIC MATERIAL WITH ENGRAVED WHITE LETTERS.
- LETTERS SHALL BE A MINIMUM 3/8" IN SIZE.
- THE LABELS SHALL BE PERMANENTLY ATTACHED TO THE APPROPRIATE PANEL.
- AC & DC CONDUIT, RACEWAY, ENCLOSURES, CABLE ASSEMBLIES AND JUNCTION BOXES SHALL BE RED BACKGROUND MATERIAL WITH WHITE LETTERING MADE OF DURABLE ADHESIVE, REFLECTIVE, WEATHER RESISTANT MATERIAL SUITABLE FOR THE ENVIRONMENT PER UL 969; TO ALERT FIRE SERVICE TO AVOID CUTTING THEM OFF.

	DATE	06/07/2024	CLIENT RESIDENCE 1234 STREET NAME, CITY, CA 9874 APN NUMBER: ----	#	REVISION/UPDATE	DATE	SIGNED & STAMP BY	E-3 SHEET NO.	PLACARDS	SHEET INFORMATION
	PROJECT ID	OPS 034-598								
	CREATED BY	MSB								
	CHECKED BY	SSN								
	SCALE	NTS								
	SHEET SIZE	11"X 17"								
	ANSI B									
HOMEOWNER INFO										

Q.PEAK DUO BLK ML-G10+ SERIES



390-410 Wp | 132 Cells
20.9 % Maximum Module Efficiency

MODEL Q.PEAK DUO BLK ML-G10+/t



Breaking the 20% efficiency barrier

Q.ANTUM DUO Z Technology with zero gap cell layout boosts module efficiency up to 20.9%.



A reliable investment

Inclusive 25-year product warranty and 25-year linear performance warranty¹.



Enduring high performance

Long-term yield security with Anti LeTID Technology, Anti PID Technology² and Hot-Spot Protect.



Extreme weather rating

High-tech aluminium alloy frame, certified for high snow (5400 Pa) and wind loads (4000 Pa).



Innovative all-weather technology

Optimal yields, whatever the weather with excellent low-light and temperature behaviour.



The most thorough testing programme in the industry

Qcells is the first solar module manufacturer to pass the most comprehensive quality programme in the industry: The new "Quality Controlled PV" of the independent certification institute TÜV Rheinland.

¹ See data sheet on rear for further information.

² APT test conditions according to IEC/TS 62804-4:2015, method A (-1500V, 96h)

The ideal solution for:



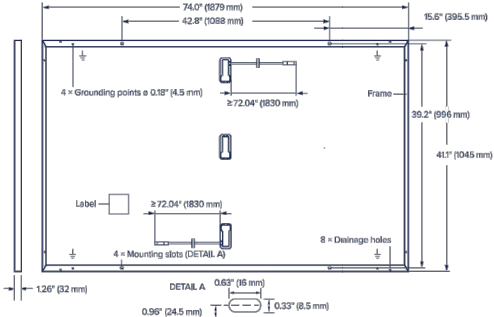
Rooftop arrays on residential buildings



Q.PEAK DUO BLK ML-G10+ SERIES

Mechanical Specification

Format	74.0 in × 41.1 in × 1.26 in (including frame) (1879 mm × 1045 mm × 32 mm)
Weight	48.5 lbs (22.0 kg)
Front Cover	0.13 in (3.2 mm) thermally pre-stressed glass with anti-reflection technology
Back Cover	Transparent composite film with black grid
Frame	Black anodised aluminium
Cell	6 × 22 monocrystalline Q.ANTUM solar half cells
Junction box	2.09-3.98 in × 1.26-2.36 in × 0.59-0.71 in (53-101 mm × 32-60 mm × 15-18 mm), IP67, with bypass diodes
Cable	4 mm ² Solar cable; (+) ≥ 72.04 in (1830 mm), (-) ≥ 72.04 in (1830 mm)
Connector	Stäubli MC4; IP68



Electrical Characteristics

POWER CLASS		390		395		400	405		410			
MINIMUM PERFORMANCE AT STANDARD TEST CONDITIONS, STC ¹ (POWER TOLERANCE +5W/-0 W)												
Minimum	Power at MPP ¹	P _{MPP} [W]	390	BSTC*		BSTC*	BSTC*		BSTC*		BSTC*	
			426.6		395	432.1	400	437.5	405	443.0	410	448.5
	Short Circuit Current ¹	I _{SC} [A]	11.01	12.05	11.04	12.08	11.07	12.11	11.10	12.15	11.13	12.18
	Open Circuit Voltage ¹	V _{OC} [V]	45.49	45.65	45.52	45.68	45.55	45.72	45.59	45.75	45.62	45.78
	Current at MPP	I _{MPP} [A]	10.39	11.37	10.45	11.43	10.50	11.49	10.56	11.55	10.61	11.61
	Voltage at MPP	V _{MPP} [V]	37.54	37.53	37.81	37.81	38.09	38.08	38.36	38.35	38.63	38.62
Efficiency ¹	η [%]	≥19.9		≥20.1		≥20.4		≥20.6		≥20.9		

Bifaciality of P_{MPP} and I_{SC} 70 % ± 10 % • Bifaciality given for rear side irradiation on top of STC (front side) • According to IEC 60904-1 according to IEC 60904-3

¹ Measurement tolerances P_{MPP} ± 3 %; I_{SC} V_{OC} ± 5 % at STC: 1000 W/m²; *at BSTC: 1000 W/m² + φ × 135 W/m², φ = 70 % ± 10 %, 25 ± 2 °C, AM 1.5 according to IEC 60904-3

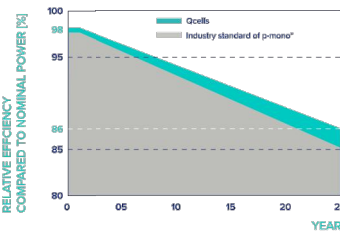
MINIMUM PERFORMANCE AT NORMAL OPERATING CONDITIONS, NMOT²

Power at MPP	P _{MPP} [W]	292.6	296.3	300.1	303.8	307.6
Short Circuit Current	I _{SC} [A]	8.87	8.89	8.92	8.94	8.97
Open Circuit Voltage	V _{OC} [V]	42.90	42.93	42.96	42.99	43.03
Current at MPP	I _{MPP} [A]	8.16	8.21	8.26	8.31	8.36
Voltage at MPP	V _{MPP} [V]	35.86	36.10	36.33	36.57	36.80

¹ Measurement tolerances P_{MPP} ± 3 %; I_{SC} V_{OC} ± 5 % at STC: 1000 W/m², 25 ± 2 °C, AM 1.5 according to IEC 60904-3 • ² 800 W/m², NMOT, spectrum AM 1.5

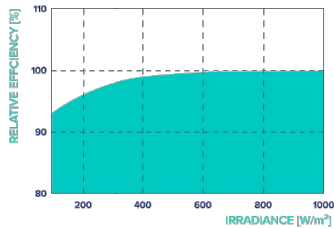
Qcells PERFORMANCE WARRANTY

PERFORMANCE AT LOW IRRADIANCE



At least 98 % of nominal power during first year. Thereafter max. 0.5 % degradation per year. At least 93.5 % of nominal power up to 10 years. At least 86 % of nominal power up to 25 years.

All data within measurement tolerances. Full warranties in accordance with the warranty terms of the Qcells sales organisation of your respective country.



Typical module performance under low irradiance conditions in comparison to STC conditions (25 °C, 1000 W/m²).

TEMPERATURE COEFFICIENTS

Temperature Coefficient of I _{SC}	α	[%/K]	+0.04	Temperature Coefficient of V _{OC}	β	[%/K]	-0.27
Temperature Coefficient of P _{MPP}	γ	[%/K]	-0.34	Nominal Module Operating Temperature	NMOT	[°F]	109 ± 5.4 (43 ± 3 °C)

Properties for System Design

Maximum System Voltage	V _{SYS} [V]	1000 (IEC)/1000 (UL)	PV module classification	Class II
Maximum Series Fuse Rating	[A DC]	20	Fire Rating based on ANSI/UL 61730	TYPE 2
Max. Design Load, Push/Pull ³	[lbs/ft ²]	75 (3600 Pa)/55 (2660 Pa)	Permitted Module Temperature on Continuous Duty	-40 °F up to +185 °F (-40 °C up to +85 °C)
Max. Test Load, Push/Pull ³	[lbs/ft ²]	113 (5400 Pa)/84 (4000 Pa)		

³ See Installation Manual

Qualifications and Certificates

UL 61730, CE-compliant,
Quality Controlled PV - TÜV Rheinland,
IEC 61215:2016, IEC 61730:2016,
U.S. Patent No. 9,893,215 (solar cells),



Qcells pursues minimizing paper output in consideration of the global environment.

Note: Installation instructions must be followed. Contact our technical service for further information on approved installation of this product.
Hanwha Q CELLS America Inc. 400 Spectrum Center Drive, Suite 1400, Irvine, CA 92618, USA | TEL: +1 949 748 59 96 | EMAIL: hqc-inquiry@qcells.com | WEB: www.qcells.com

qcells

Single Phase Energy Hub Inverter with Prism Technology

For North America

SE3000H-US / SE3800H-US / SE6000H-US / SE7600H-US / SE10000H-US / SE11400H-US⁽¹⁾



HOME BACKUP

Optimized battery storage with HD-Wave technology

- Record-breaking 99% weighted efficiency with 200% DC oversizing
- Multi-inverter, scalable storage solution
- With enhanced battery power up to 10kW
- Small, lightweight, and easy to install
- Integrated arc fault protection and rapid shutdown for NEC 2014, NEC 2017 and NEC 2020, per article 690.11 and 690.12
- Modular design, future ready with optional upgrades to:
 - DC-coupled storage for full or partial home backup
 - Built-in consumption monitoring
 - Direct connection to the SolarEdge smart EV charger
- Embedded revenue grade production data, ANSI C12.20 Class 0.5

Single Phase Energy Hub Inverter with Prism Technology

For North America

SE3000H-US / SE3800H-US / SE6000H-US / SE7600H-US / SE10000H-US / SE11400H-US⁽¹⁾

	SE3000H-US	SE3800H-US	SE6000H-US	SE7600H-US	SE10000H-US	SE11400H-US	UNITS
OUTPUT - AC ON GRID							
Rated AC Power	3000	3800 @ 240V 3300 @ 208V	6000 @ 240V 5000 @ 208V	7600	10000	11400 @ 240V 10000 @ 208V	W
Maximum AC Power Output	3000	3800 @ 240V 3300 @ 208V	6000 @ 240V 5000 @ 208V	7600	10000	11400 @ 240V 10000 @ 208V	W
AC Frequency Range (min - nom - max)			59.3 - 60 - 60.5 ⁽²⁾				Hz
Maximum Continuous Output Current @ 240V	12.5	16	25	32	42	47.5	A
Maximum Continuous Output Current @ 208V	-	16	24	-	-	48.5	A
GFDI Threshold			1				A
Total Harmonic Distortion (THD)			<3				%
Power Factor			1, adjustable -0.85 to 0.85				
Utility Monitoring/Islanding/Protection/Country Configurable/Thresholds			Yes				
Charge Battery from AC (if allowed)			Yes				
Typical Nighttime Power Consumption			<2.5				W
OUTPUT - AC BACKUP ⁽³⁾							
Rated AC Power in Backup Operation ⁽⁴⁾	3000	3800 7600*	6000	7600 10300*	10000	10300	W
AC L-L Output Voltage Range in Backup			211 - 264				Vac
AC L-N Output Voltage Range in Backup			105 - 132				Vac
AC Frequency Range in Backup (min - nom - max)			55 - 60 - 65				Hz
Maximum Continuous Output Current in Backup Operation	12.5	16 32*	25	32 43*	42	43	A
GFDI			1				A
THD			<5				%
OUTPUT - SMART EV CHARGER AC							
Rated AC Power			9600				W
AC Output Voltage Range			211 - 264				Vac
On-Grid AC Frequency Range (min - nom - max)			59.3 - 60 - 60.5				Hz
Maximum Continuous Output Current @ 240V (grid, PV and battery)			40				Aac
INPUT - DC (PV AND BATTERY)							
Transformer-less, Ungrounded			Yes				
Max Input Voltage			480				Vdc
Nom DC Input Voltage			380				Vdc
Reverse-Polarity Protection			Yes				
Ground-Fault Isolation Detection			600kΩ Sensitivity				
INPUT - DC (PV)							
Maximum DC Power @ 240V	6000	7600 15200*	12000	15200 22800*	22000	22800	W
Maximum DC Power @ 208V	-	6600	10000	-	-	20000	W
Maximum Input Current ⁽⁵⁾ @ 240V	8.5	10.5 20*	16.5	20 31*	27	31	Adc
Max. Input Short Circuit Current		9	13.5	-	-	27	Adc
Maximum Inverter Efficiency	99			99.2			%
CEC Weighted Efficiency			99			99 @ 240V 98.5 @ 208V	%
2-pole Disconnection			Yes				

* Supported with PN SExxxxH-USMxxxxxx or SExxxxH-USMxxxxxx
(1) These specifications apply to inverters with part numbers SExxxxH-USMxxxxxx or SExxxxH-USNxxxxxx and connection unit model number DCD-1PH-US-P4H-F-x
(2) For other regional settings please contact SolarEdge support
(3) Not designed for standalone applications and requires AC for commissioning. Backup functionality is only supported for 240V grid
(4) Rated AC power in Backup Operation are valid for installations with multiple inverters. For a single backup inverter operation, rated AC power in Backup is 90% of the value stated
(5) A higher current source may be used; the inverter will limit its input current to the values stated

Single Phase Energy Hub Inverter with Prism Technology

For North America

SE3000H-US / SE3800H-US / SE6000H-US / SE7600H-US / SE10000H-US / SE11400H-US⁽¹⁾

	SE3000H-US	SE3800H-US	SE6000H-US	SE7600H-US	SE10000H-US	SE11400H-US	UNITS
INPUT – DC (BATTERY)							
Supported Battery Types	SolarEdge Energy Bank, LG RESU Prime ⁽⁶⁾						
Number of Batteries per Inverter	Up to 3 SolarEdge Energy Bank, up to 2 LG RESU Prime						
Continuous Power ⁽⁷⁾	6000	7600	10000				W
Peak Power ⁽⁸⁾	6000	7600	10000				W
Max Input Current	16	20	26.5				Adc
2-pole Disconnection	Yes						
SMART ENERGY CAPABILITIES							
Consumption Metering	Built – in ⁽⁹⁾						
Backup & Battery Storage	With Backup Interface (purchased separately) for service up to 200A; Up to 3 inverters						
EV Charging	Direct connection to Smart EV charger						
ADDITIONAL FEATURES							
Supported Communication Interfaces	RS485, Ethernet, Cellular ⁽⁹⁾ , Wi-Fi (optional), SolarEdge Energy Net (optional)						
Revenue Grade Metering, ANSI C12.20	Built – in ⁽⁹⁾						
Integrated AC, DC and Communication Connection Unit	Yes						
Inverter Commissioning	With the SetApp mobile application using built-in Wi-Fi Access Point for local connection						
DC Voltage Rapid Shutdown (PV and Battery)	Yes, according to NEC 2014, NEC 2017 and NEC 2020 690.12						
STANDARD COMPLIANCE							
Safety	UL1741, UL1741 SA, UL1741 PCS, UL16998, UL1998, UL9540, CSA 22.2						
Grid Connection Standards	IEEE1547, Rule 21, Rule 14H						
Emissions	FCC part 15 class B						
INSTALLATION SPECIFICATIONS							
AC Output and EV AC Output Conduit Size / AWG Range	1" maximum / 14-4 AWG						
DC Input (PV and Battery) Conduit Size / AWG Range	1" maximum / 14-6 AWG						
Dimensions with Connection Unit (H x W x D)	17.7 x 14.6 x 6.8 / 450 x 370 x 174		17.7 x 14.6 x 6.8 / 450 x 370 x 174 17.7 x 14.6 x 6.8 / 450 x 370 x 174*		17.7 x 14.6 x 6.8 / 450 x 370 x 174		in / mm
Weight with Connection Unit	26 / 11.8		26 / 11.8 41.7 / 18.9*		41.7 / 18.9		lb / kg
Noise	< 25	< 25 * < 50*	< 25	< 50			dBA
Cooling	Natural Convection						
Operating Temperature Range	-40 to +140 / -40 to +60 ⁽¹⁰⁾						°F / °C
Protection Rating	NEMA 4						

(6) The part numbers SExxxxH-USMxxxxxx only support the SolarEdge Energy Bank. The part numbers SExxxxH-USNxxxxxx support both SolarEdge Energy Bank and LG RESU Prime batteries
(7) Requires supporting inverter firmware
(8) For consumption metering current transformers should be ordered separately: SECT-SPL-225A-T-20 or SEACT0750-400NA-20 units per box. Revenue grade metering is only for production metering
(9) Information concerning the Data Plan's terms & conditions is available in the following link:
<https://www.solaredge.com/sites/default/files/se-communication-plan-terms-and-conditions-eng.pdf>
(10) Full power up to at least 50°C / 122°F; for power de-rating information refer to: <https://www.solaredge.com/sites/default/files/se-temperature-derating-note-na.pdf>

Power Optimizer

For North America

S440 S500



POWER OPTIMIZER

PV power optimization at the module level

- Specifically designed to work with SolarEdge residential inverters
- Detected abnormal PV connector behavior, preventing potential safety issues*
- Module-level voltage shutdown for installer and firefighter safety
- Superior efficiency (99.5%)
- Mitigates all types of module mismatch loss, from manufacturing tolerance to partial shading
- Faster installations with simplified cable management and easy assembly using a single bolt
- Flexible system design for maximum space utilization
- Compatible with bifacial PV modules
- Meets NEC requirements for arc fault protection (AFCI) and Photovoltaic Rapid Shutdown System (PVRSS)

* Expected availability in 2022

[solaredge.com](https://www.solaredge.com)

solaredge

Power Optimizer

For North America

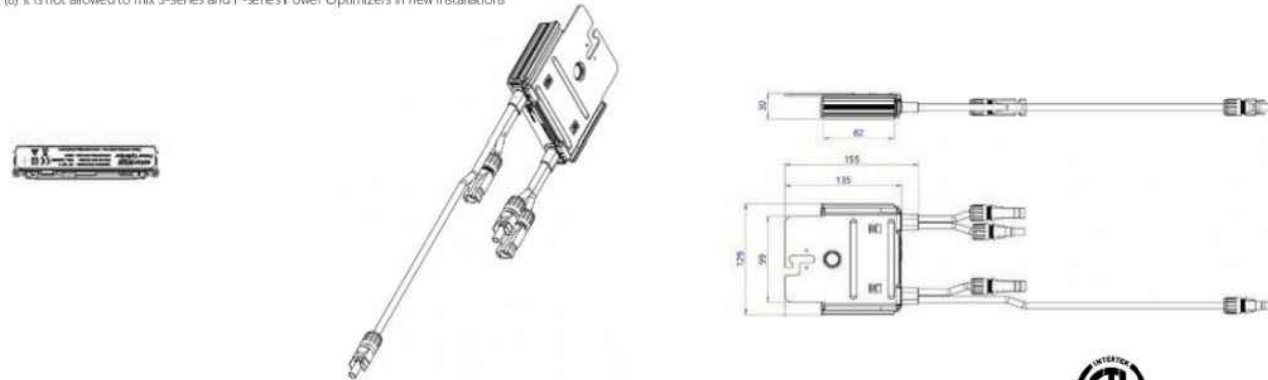
S440 S500

	S440	S500	Unit
INPUT			
Rated Input DC Power ^m	440	500	W
Absolute Maximum Input Voltage (Voc)	60		Vdc
MPPT Operating Range	8 - 60		Vdc
Maximum Short Circuit Current (Isc) of Connected PV Module	14.5	15	Adc
Maximum Efficiency	99.5		%
Weighted Efficiency	98.6		%
Overtoltage Category	II		
OUTPUT DURING OPERATION			
Maximum Output Current	15		Adc
Maximum Output Voltage	60		Vdc
OUTPUT DURING STANDBY (POWER OPTIMIZER DISCONNECTED FROM INVERTER OR INVERTER OFF)			
Safety Output Voltage per Power Optimizer	1+/-0.1		Vdc
STANDARD COMPLIANCE			
Photovoltaic Rapid Shutdown System	NEC 2014, 2017 & 2020		
EMC	FCC Part 15 Class B, IEC61000-6-2, IEC61000-6-3		
Safety	IEC62109-1 (class II safety), UL1741		
Material	UL94 V-0, UV Resistant		
RoHS	Yes		
Fire Safety	VDE-AR-E 2100-712:2013-05		
INSTALLATION SPECIFICATIONS			
Maximum Allowed System Voltage	1000		Vdc
Dimensions (W x L x H)	129 x 153 x 30 / 5.07 x 6.02 x 1.18		mm / in
Weight (including cables)	655 / 1.5		gr / lb
Input Connector	MC4 ^m		
Input Wire Length	0.1 / 0.32		m / ft
Output Connector	MC4		
Output Wire Length	(+) 2.3, (-) 0.10 / (+) 7.54, (-) 0.32		m / ft
Operating Temperature Range ^m	-40 to +85		°C
Protection Rating	IP68 / NEMA6P		
Relative Humidity	0 - 100		%

(1) Rated power of the module at STC will not exceed the power optimizer Rated Input DC Power. Modules with up to +5% power tolerance are allowed.
(2) For other connector types please contact SolarEdge.
(3) For ambient temperature above +70°C / +158°F power de-rating is applied. Refer to Power Optimizers Temperature De-Rating Technical Note for more details.

PV System Design Using a SolarEdge Inverter		Single Phase HD-Wave	Three Phase for 208V Grid	Three Phase for 277/480V Grid	
Minimum String Length (Power Optimizers)	S440, S500	8	10	18	
Maximum String Length (Power Optimizers)		25		50 ⁽⁴⁾	
Maximum Nominal Power per String		5700 (6000 with SE7600-US-SE11400-U)	6000	12750	W
Maximum Allowed Connected Power per String ⁽⁵⁾ (Permitted only when the difference in connected power between strings is 1000W or less)		Refer to Footnote 5	One string 7200 Two strings or more 7800	15000	W
Parallel Strings of Different Lengths or Orientations			Yes		

(4) A string with more than 30 optimizers does not meet NEC rapid shutdown requirements; safety voltage will be above the 30V requirement.
(5) If the inverters rated AC power < maximum nominal power per string, then the maximum power per string will be able to reach up to the inverters maximum input DC power. Refer to: <https://www.solaredge.com/sites/default/files/se-power-optimizer-single-string-design-application-note.pdf>
(6) It is not allowed to mix S-series and P-series Power Optimizers in new installations.



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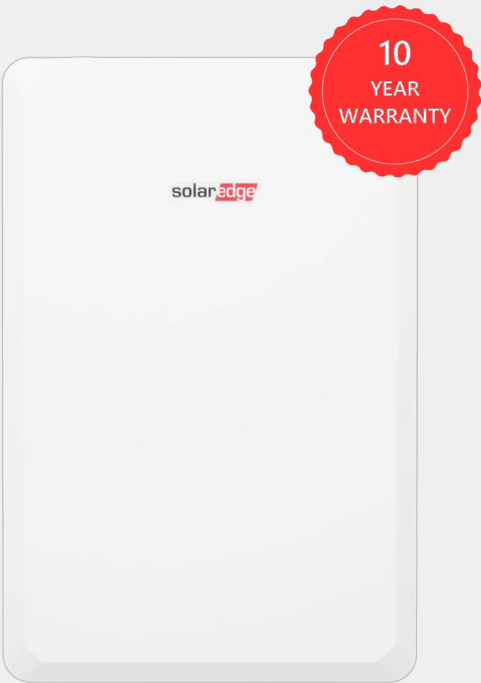


RoHS

SolarEdge Home Battery

For North America

BAT-10K1P



BATTERIES

Optimized for SolarEdge Home Hub Inverters⁽¹⁾

- DC coupled battery featuring superior overall system efficiency, generating more energy to store and use for on-grid and backup power applications
- Integrates seamlessly with the complete SolarEdge Home ecosystem using SolarEdge Home Network, offering a single source for warranty, support, and training, to streamline logistics & operations
- Enhanced safety with Rapid Shutdown, SafeDC™
- Qualified by the latest and most stringent UL fire safety standard
- Solar, storage, EV charging, and smart devices all monitored and managed by a single app to an optimized production, consumption, and backup* power
- Simple plug and play installation, with automatic SetApp-based configuration
- Flexible installation - wall or floor mount, all indoor or outdoor
- Wireless communication to the inverter, reducing wiring, labor, and installation

* Backup applications are subject to local regulation and may require additional components and firmware upgrade

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SolarEdge Home Battery

For North America

BAT-10K1P

BAT-10K1PS0B-02			
BATTERY SPECIFICATION			
Usable Energy (100% depth of discharge)		9700	Wh
Continuous Output Power		5000	W
Peak Output Power (for 10 seconds)		7500	W
Peak Roundtrip Efficiency		>94.5	%
Warranty ⁽¹⁾		10	Years
Voltage Range		350-450	Vdc
Communication Interfaces		Wireless and RS485 ⁽²⁾	
Batteries per Inverter		Up to 3 ⁽³⁾	
STANDARD COMPLIANCE			
Certification	Cell	UL1642	
	Battery	UL1973, UL9540A, UL9540, UN38.3	
Emissions		FCC Part 15 Class B	
MECHANICAL SPECIFICATIONS			
Dimensions (W x H x D)		31.1 x 46.4 x 9.84 / 790 x 1179 x 250	in / mm
Weight		262 / 119	lb / kg
Mounting		Floor ⁽⁴⁾ or wall mount ⁽⁵⁾	
Operating Temperature ⁽⁶⁾		+14 to +122 / -10 to +50	°F / °C
Storage Temperature (more than 3 months)		+14 to +86 / -10 to +30	°F / °C
Storage Temperature (less than 3 months)		-22 to + 140 / -30 to +60	°F / °C
Altitude		6562 / 2000	ft / m
Enclosure Protection		IP55 / NEMA 3R - indoor and outdoor (water and dust protection)	
Cooling		Natural convection	
Noise (at 1m distance)		<25	dBA

(1) For warranty details, see the SolarEdge Home Battery Limited Product Warranty.
(2) The SolarEdge Home Battery is designed for use with SolarEdge Home Network for wireless communication. The inverter might require a matching SolarEdge Home Network Plug-in (more details below). Using RS485 could reduce the usable energy to 9500Wh.
(3) Installations with multiple SolarEdge Home Battery connected to a single inverter require a pair of branch connectors (DC + and DC -) per battery excluding the last battery. The branch connectors should be purchased separately.
(4) The floor stand is purchased separately. One floor stand is required per SolarEdge Home Battery. See the Accessories P/N table below.
(5) Wall mount installation requires handles that should be purchased separately. See the Accessories P/N table below.
(6) Note that operating the SolarEdge Home Batteries at extreme temperatures for extended durations of time may void the SolarEdge Home Battery warranty coverage. Please see the SolarEdge Home Battery Limited Product Warranty for additional details.

SolarEdge Home Battery – Accessories (purchased separately)	
ACCESSORY	P/N
Floor stand	IAC-RBAT-FLRSTD-01
Branch connectors set (includes a pair of DC + and DC - connectors) Required for installations with multiple SolarEdge Home Batteries with a single inverter	IAC-RBAT-USYCBL-01
Handles	IAC-RBAT-HANDLE-01
SolarEdge Home Network Plug-in	ENET-HBNP-01

SolarEdge Home Backup Interface

For North America

BI-EUSGN-02 / BI-NUSGN-02

HOME BACKUP



Backup Interface for Flexible Backup

- Automatically provides backup power to home loads in the event of grid interruption
- Full flexibility in which loads to back up – the entire home or selected loads
- Scalable solution to support higher power and higher capacity^(*)
- Built-in Auto Transformer and Energy Meter for easier and faster installation
- Seamless integration with the SolarEdge Home Hub Inverter to manage and monitor both PV generation and energy storage
- Generator connection support^(*)

^(*) Requires supporting inverter firmware
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SolarEdge Home Backup Interface

For North America

BI-EUSGN-02 / BI-NUSGN-02

	BI-EUSGN-02	BI-NUSGN-02	
INPUT FROM GRID			
AC Current Input	200		A
AC Output Voltage (Nominal)	240		Vac
AC Output Voltage Range	211 – 264		Vac
AC Frequency (Nominal)	60		Hz
AC Frequency Range	59.3 – 60.5		Hz
Microgrid Interconnection Device Rated Current	200		A
Service Side AC Main Circuit Breaker Rated Current	200	N/A	A
Service Side AC Main Circuit Breaker Interrupt Current	10000	N/A	A
Grid Disconnection Switchover Time	<100		ms
OUTPUT TO MAIN DISTRIBUTION PANEL			
Maximum AC Current Output	200		A
AC L-L Output Voltage (Nominal)	240		Vac
AC L-L Output Voltage Range	211 – 264		Vac
AC Frequency (Nominal)	60		Hz
AC Frequency Range	59.3 – 60.5		Hz
Maximum Inverters AC Current Output in Backup Operation	144		A
Imbalance Compensation in Backup Operation	5000		W
AC L-N Output Voltage in Backup (Nominal)	120		V
AC L-N Output Voltage Range in Backup	105 – 132		V
AC Frequency Range in Backup	55 – 65		Hz
INPUT FROM INVERTER			
Number of Inverter Inputs	Up to 3		#
Maximum Rated AC Power in On-Grid and Backup Operation	11400		W
Maximum Continuous Current in On-Grid and Backup Operation	48		A
Factory Installed Inverter Input AC Circuit Breaker	40		A
Upgradability	Up to 3 x 63A CB ⁽¹⁾		
GENERATOR ⁽²⁾			
Maximum Rated AC Power	22500		W
Maximum Continuous Input Current	94		Adc
Dry Contact Switch Voltage Rating	250 / 30		Vac / Vdc
Dry Contact Switch Current Rating	5		A
2-wire Start Switch	Yes		
ADDITIONAL FEATURES			
Installation Type	Suitable for use as service equipment	For main lug only	
Number of Communication Inputs	2		
Communication	RS485		
Energy Meter (for Import/Export)	1% accuracy		
Manual Control Over Microgrid Interconnection Device	Yes		

⁽¹⁾ Each 40A CB supports up to one 7.6kW Inverter, with each 63A CB supporting one 10kW and one 11.4kW Inverter. The CB upgrade kit is available with the following part numbers: for 40A CB, CB-UPG-40-01; for 63A, CB CB-UPG-63-01. 20A, 30A and 50A breakers can be used for inverters with lower power ratings.
⁽²⁾ Requires supporting inverter firmware.

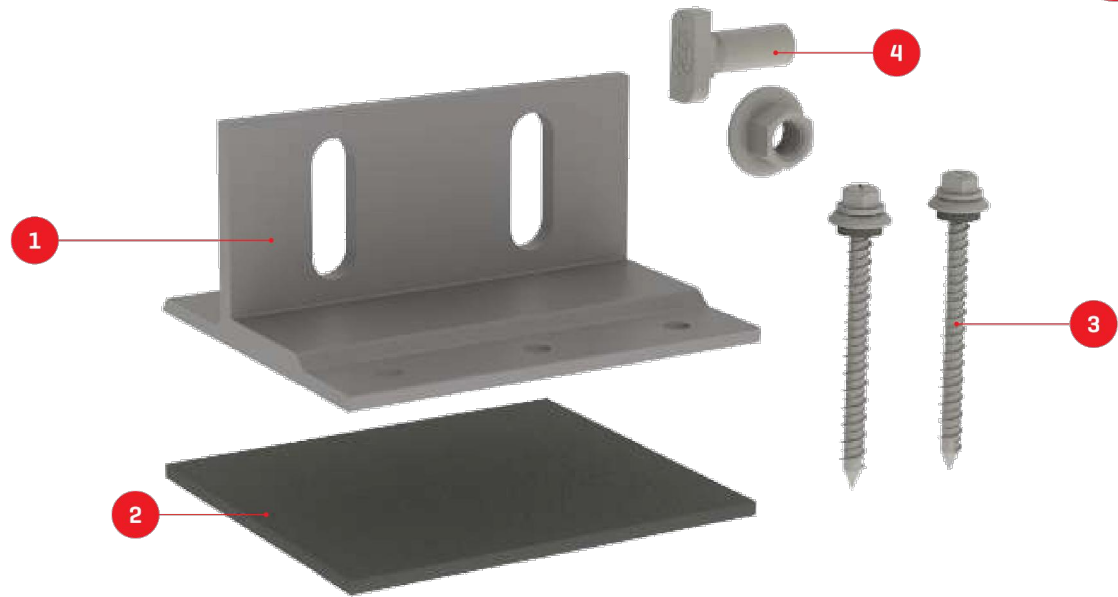
SolarEdge Home Backup Interface

For North America

BI-EUSGN-02 / BI-NUSGN-02

	BI-EUSGN-02	BI-NUSGN-02	
STANDARD COMPLIANCE			
Safety	UL1741, CSA 22.2 NO. 107		
Emissions	UL869A	N/A	
	FCC part 15 class B		
INSTALLATION SPECIFICATIONS			
Supported Inverters	StorEdge single phase inverter, SolarEdge Home Hub Inverter		
AC From Grid Conduit Size / AWG Range	2" conduit / 0 AWG – 4/0		
AC Inverter Conduit Size / AWG Range	1" conduit / 14 – 4 AWG		
AC Generator Input Conduit Size / AWG Range	1" conduit / 8 – 3 AWG		
Communication Conduit Size / AWG Range	3/4" / 24 – 10 AWG		
Weight	73 / 33		lb / Kg
Cooling	Fan (user replaceable)		
Noise	< 50		dBA
Operating Temperature Range	-40 to +122 / -40 to +50		°F / °C
Protection Rating	NEMA 3R, IP44		
Dimensions (HxWxD)	20.59 x 13.88 x 8.62 / 523.5 x 352.5 x 219		in / mm

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Splice Foot X

Patent Pending

TECHNICAL SHEET

Item Number	Description	Part Number
1	Splice Foot X	4000113 Splice Foot X Kit, Mill
2	K2 EverSeal-	
3	M5 x 60 lag screws	
4	T-Bolt & Hex Nut Set	

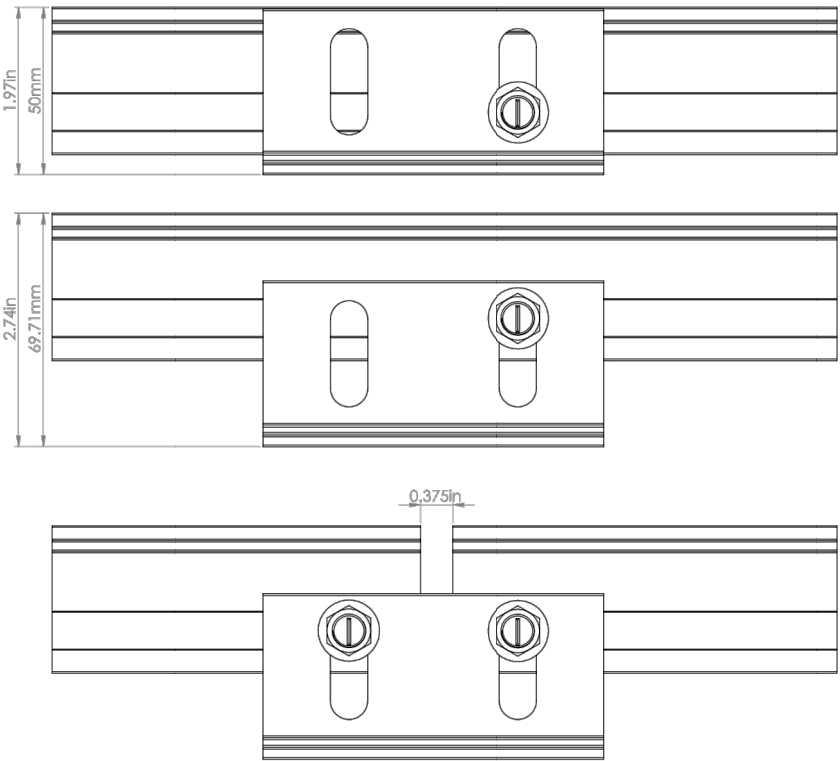
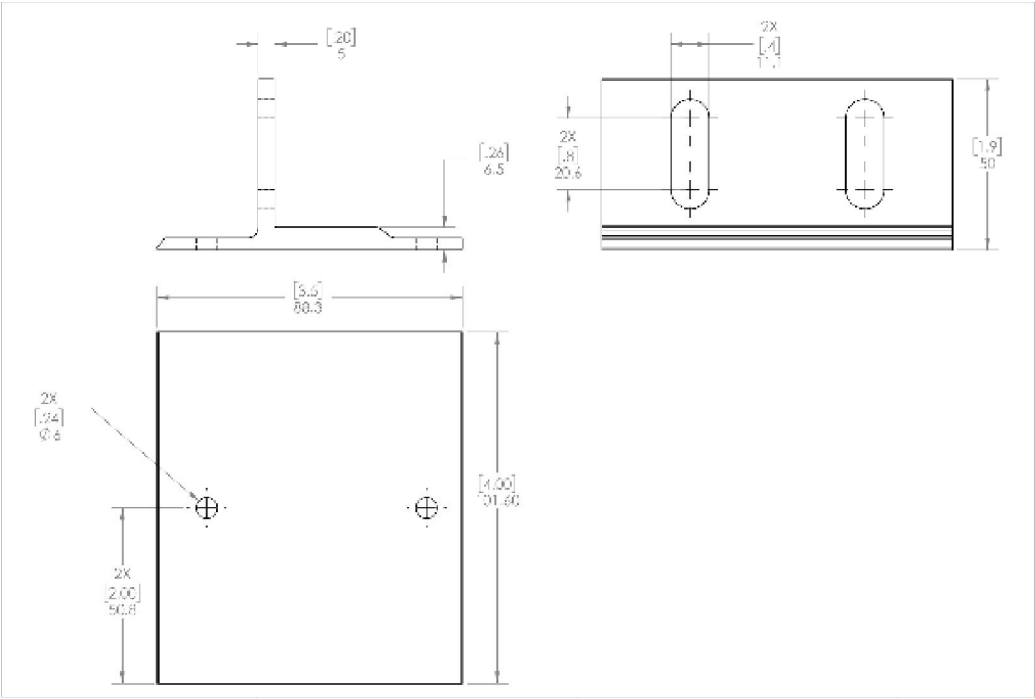
Technical Data

	Splice Foot X
Roof Type	Composition shingle
Material	Aluminum with stainless steel hardware
Finish	Mill
Roof Connection	M5 x 60 lag screws
Code Compliance	UL 2703
Compatibility	CrossRail 44-X, 48-X, 48-XL, 80

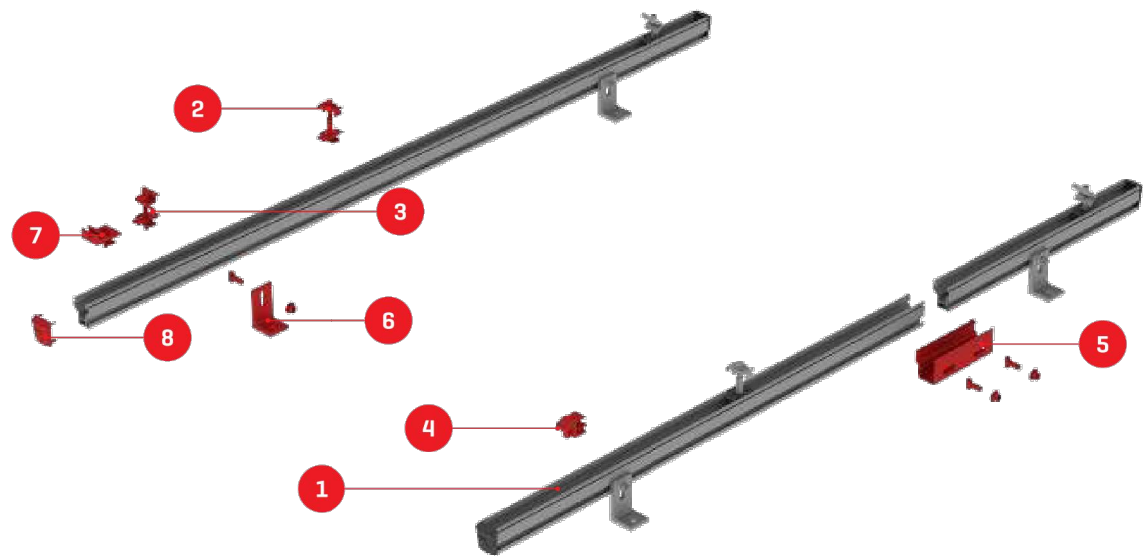
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Units: [in] mm



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CrossRail System

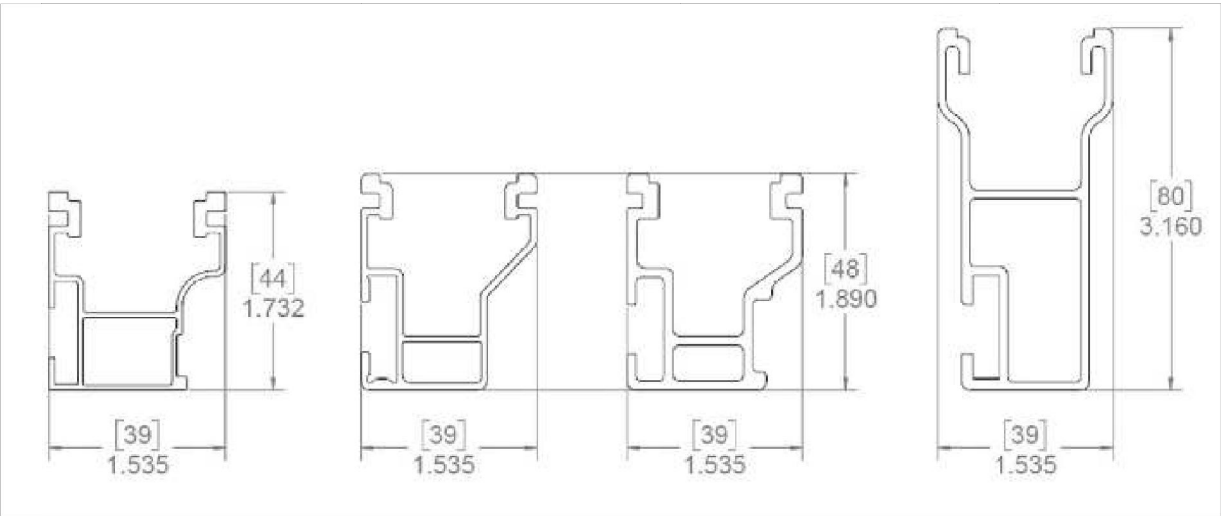
TECHNICAL SHEET

Item Number	Description	Part Number
1	CrossRail 44-X [shown] all CR profiles applicable	4000019 [166" mill], 4000020 [166" dark] , 4000021 [180" mill], 4000022 [180" dark]
2	CrossRail Mid Clamp	4000601-H [mill], 4000602-H [dark]
3	CrossRail [Standard] End Clamp	4000429 [mill], 4000430 [dark]
4	Yeti Hidden End Clamp for CR	4000050-H
5	CrossRail 44-X Rail Connector [shown] CR 48-X, 48-XL Rail Connector available	4000051 [mill], 4000052 [dark]
6	L-Foot Slotted Set	4000630 [mill], 4000631 [dark]
7	Everest Ground Lug	4000006-H
8	CrossRail 44-X End Cap [shown] CrossRail 48-X, 48-XL and 80 available	4000067

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Units: [mm] in



Technical Data

	CrossRail System
Roof Type	Composition shingle, tile, standing seam
Material	High corrosion resistance stainless steel and high grade aluminum
Flexibility	Modular construction, suitable for any system size, height adjustable
PV Modules	For all common module types
Module Orientation	Portrait and landscape
Roof Connection	Drill connection into rafter
Structural Validity	IBC compliant, stamped engineering letters available for all solar states
Warranty	25 years